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USER MANUAL



MISPA HX88

Automatic Hematology Analyzer

Please carefully read and understand these statements:

1. The manual is subject to change without notice.
2. The manual is owned by Agappe Diagnostics
   No part of this manual may be copied or reproduced or disclosed to any third party in any form or by any means without written consent of Agappe Diagnostics
3. In no event will Agappe Diagnostics be liable for direct or indirect damages or damages arising of any use of the product not specified in the manual.
4. Failure to use the product according to the methods prescribed in the manual may damage the protection provided by the product.

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Foreword

Thank you for purchasing the Automatic Hematology Analyzer (hereinafter referred to as "the analyzer") manufactured by Agappe (hereinafter referred to as "the Company")

- Please use the analyzer under the supervision of technical personnel that have undergone professional training. For clinical examinations, please use.
- Use of incorrect reagents and consumables may lead to incorrect test results and may harm the analyzer or operator.
- To guarantee correct use of the analyzer, please thoroughly read the manual.
- Keep the manual handy for future reference.
- Our company could provide one of two compressor models that are called CP-10 or CP-11. The CP-10 is suggested to work under the condition of AC 220V~50Hz/60Hz, however the CP-11 is suggested to work under the condition of AC 110V-117V~60Hz. Please confirm the compressor model with the salesman before purchase.

Product Warranties

Due to space limitations, many more could have been included. However, the methods documented in this manual can ensure normal use of the Automatic Hematology Analyzer by users.

a. Warranty coverage

For any failure caused by defects in manufacturing, the Company will provide repair services free of charge. During the service, some substitutes may be used or the parts to be repaired may be replaced.

b. Disclaimers

The following is not covered under the warranty even within the warranty period:

- Failures resulting in the use of the product not in the applicable environment as specified in the manual.
- Failures resulting in the use of the product not at the specified power supply, voltage, or frequency, and those caused due to grid exception.
- Failures resulting in circuit corrosion and degradation due to highly corrosive gases in the air.
- Failures resulting in the use of hardware, software or supporting supplies not provided by the Company.
- Failures resulting in the use of the product according to a method not specified in the manual or incorrect use of the product, and those caused due to repair by a party other than the Company.
- Failures resulting in the maintenance by a party other than the maintenance company appointed by the Company.
- Failures resulting in the move and transportation of the installed product with a method not authorized by the Company.
- Failures resulting in unauthorized disassembly or alternation of the product.
- Failures resulting in fires, earthquakes, wind damages, floods, lightning strokes, crimes, violence, terrorist acts, radioactive contaminations, wars, hazardous substances, and other acts of God.
- Failures occurred to scrapped analyzers or the analyzers sub-purchased.

  without the written consent of the Company.

c. Declaration

The manual is subject to change without notice.

Installation, Handling and After-Sales Service

a. Installation and Handling

When installing, based on referring to the manual, you should prepare the hardware conditions and the corresponding working environment required for the analyzer installation.

When you need to move the analyzer after purchase, in order to avoid the failure caused by the handling, please contact the service department of the Company or an authorized dealer.

b. After-Sales Service

- For service and maintenance contracts, please contact the service department of the Company or an authorized dealer.
- For services after the warranty period, please prepare maintenance contracts, check service contracts, etc. to consult with the service department of the Company or an authorized dealer.

Storage and Shipping Conditions

Storage conditions

A packaged analyzer should be stored in a well-ventilated place without corrosive gases under -10℃ to 40℃ with the relative humidity of 10%— 90%.

Shipping Conditions

Please use the special packing box designated by the Company for shipping. It is forbidden to ship without packaging.

When shipping, the packing box must be fastened to the transportation tool according to the packing instructions. No dumping or crashing.

When transporting, prevent exposure to the Sun and rain and moisture.

Production Date and Duration

For the production date of the analyzer, please refer to its nameplate. The duration of the analyzer is five years.

The duration of the product is determined based on the aging test and risk management. During the use of the product, users need to maintain and repair the product according to the manual. When the product expires, if its basic security and effectiveness can be maintained after maintenance and repair, its continued use is allowed.

Disposal

The analyzer shall be disposed of in accordance with relevant national laws and regulations.

How to Get Help

If you have any questions when installing or using this analyzer, please contact the service department of Agappe or an authorized dealer. Authorized dealer.

- Company website: www.Agappe .com

Please prepare the following materials When seeking for help:

- The production number of your Automatic Hematology Analyzer（can be found on the nameplate at the rear side of the Analyzer)
- Description of the problem
- Methods and operations you've taken to try to resolve the problem.
- Your contact information such as the phone number and email

User Training

The Company provides operation training services for Automatic Hematology Analyzer. For more information, please consult our service department.

Vigilance

If any serious incident has occurred in relation to this product, please contact the manufacturer and report to the local competent authority.

Precautions for Installation and Maintenance

Users must not attempt to install the analyzer themselves. To ensure the safety and normal use of the analyzer, our service department will provide you with installation services.
The items that users can maintain themselves are limited to the scope described in the manual. Learn more about it before maintenance. Do not attempt maintenance items other than those specified in the manual, otherwise the analyzer will be dangerous and may cause personal injury.
If the warning signs are worn or damaged, please contact our service.
department for replacement.

Precautions Before Operation

Operation and instructions for use of the Analyzer

Protective clothing is especially important during maintenance and inspections. Before using the analyzer, be sure to wear protective clothing to prevent contamination by infectious substances. If there are specific regulations or instructions, please follow them.
This analyzer is a Class 1 laser product. Do not perform any operations other than those specified in the manual. If there are any problems with the analyzer, please contact our service department.
Always follow the precautions and instructions described on the analyzer and in the manual. Failure to do so will result in personal injury and damage to the analyzer.
Not all accidents can be predicted, so the danger warning signs in this manual and on the analyzer cannot cover all possible situations. It is not enough to follow the given instructions for operation. Please be vigilant and use your common sense.

Operations of the Analyzer

When checking reagent stability and analyzer operation, please measure the controls to ensure accuracy control. Before measuring patient samples, ensure that the control results meet the requirements.
For the storage before unpacking and after unpacking, handling and use of reagents and controls, please follow the requirements in their manuals.

Other Safety Precautions

Do not modify the analyzer or use parts other than those specified, otherwise danger may occur.

Precautions for Use

In use, pay attention to abnormal sounds, liquid leaks, or other abnormal conditions. If a problem occurs, take appropriate security measures, and contact our service department.
Do not open any cover of the analyzer during use, otherwise danger may occur.

Other Considerations

(1). Environmental Conditions

The environment in which the analyzer is used must comply with the following requirements:

a. Clean air and good ventilation
b. Avoid direct sunlight.
c. The table is horizontal（slope is 1/200 or lower), and the table size is not less than 700 * 700（length width);

e. Room temperature is in the range of 15℃ to 30℃

f. Relative humidity is in the range of 30% to 85%, and no condensation will appear on the analyzer surface.

g. There must be no vibrations that the human body can feel.

h. here is an electric panel within 5 meters of the analyzer.

i. The voltage fluctuation is within ± 10%.

j. Keep away from equipment that can generate electromagnetic and strong magnetic interference （such as centrifuges, electric spark machines, mobile phones, transceivers, cordless phones, etc.）; .

k. There is a ground terminal （ground resistance is 4Ω or less)

l. When the analyzer is running, the noise is not more than 70db（excluding occasional noise and failure alarm sound);

m. When the analyzer is in the ready state, the noise is not more than 60db.

(2). Disposal of Reagents, Samples and Consumables

a. They should be disposed in accordance with the corresponding measures of national hazardous safety guidelines or regulations, or in accordance with relevant environmental protection regulations in the country of sale.

b. It is the user's responsibility to handle, store or discard the used samples in accordance with predetermined standards. Therefore, it is recommended to extract and store some samples in advance to prevent accidental loss of samples due to operating errors or other reasons.

Disposal of the Analyzer and Waste

Before disposing of the analyzer, please contact our service department to inquire about safe disposal procedures.
Dispose of the analyzer in accordance with local environmental protection regulations.
Dispose of waste in strict accordance with environmental protection regulations.

Operational Precautions

To use the Automatic Hematology Analyzer correctly, pay attention to the following matters:

Precautions for Ensuring Measurement Accuracy

- During the measurement, do not open the cover of the detection module in the analyzer, otherwise the measurement results will be inaccurate.
- Please replace the wearing parts in time, otherwise the measurement results will be inaccurate.
- Clean the sample probe regularly to ensure that the surface is not dirty, otherwise it will cause the analyzer to malfunction or the measurement results to be inaccurate.
- Never use expired reagents, calibrators, and controls. sing expired products will cause measurement errors and result in inaccurate results.
- Do not leave the sample, calibrator, or control in the sample tube for a long time, otherwise it will be concentrated due to evaporation, and accurate analysis cannot be performed, resulting in inaccurate measurement results.
- Use a standard sample tube, otherwise the measurement results may be inaccurate due to inaccurate sample aspiration.
- Do not add more reagent, otherwise cross-contamination may occur, resulting in inaccurate measurement results.
- During the measurement, do not open the cover of the working compartment, otherwise the analyzer will be faulty or the measurement results will be inaccurate.
- The fluidic system must be primed before any experiment every day, otherwise the measurement results will be inaccurate.

Precautions to Prevent the Analyzer from Damage

- Use standard sample tubes, otherwise the sampling system may be damaged.
- If liquid spills from the wiper of the sample probe, it may cause the analyzer failure such as damage to the circuit board or components.
- Never place samples or reagents on the table where the analyzer is put. Spilling samples or reagents on the Analyzer can cause the analyzer failure such as damage to circuit boards or components.

Precautions for Powering On The Analyzer

- During the maintenance of the analyzer, do not switch on or off the power supply, otherwise personal injury or damage to the analyzer may result.

Precautions for Starting Measurement

- Ensure that no one is performing maintenance, and then operate the software to start the measurement. If the measurement is started during the analyzer maintenance process, the parts or tools used for maintenance will touch the running stations and cause the analyzer to be damaged.

Data Backup

- If the analyzer fails or operates incorrectly, you cannot view or use the measurement results or parameters. For safety reasons, you are recommended to regularly back up your results and parameters.

Computer Antivirus Protection

- If the program/data is suddenly damaged or the operation/interface suddenly comes with unexpected situation, your computer may be infected with a computer virus.
- A computer virus refers to a malicious program that sneaks into your computer and can cause data anomalies or damage.
- Programs designed to protect against computer viruses are called antivirus programs.
- Possible causes of virus infection are:
  A program with a virus was downloaded during communication.

Precautions:

- Once a computer is infected with a virus, the virus may spread to other computers through communication or storage media.
- Never use programs or storage media with a suspected virus.
- Before using removable storage media, apply an antivirus program to check it for viruses.

Power Outage

- If the power supply is interrupted or a lightning strike occurs, the analyzer operating unit may malfunction, and system software, application software, and data may also be damaged.

Control Unit

- In the process of accessing the hard disk or other removable media, if the control unit is powered off suddenly, the data/software stored in it may be damaged. For how to properly turn off the control unit, refer to the relevant section of the manual.
- Do not install other programs into the control unit or change the settings in the control unit.
- Do not remove the removable media while the indicator of the removable media is on. Otherwise, the analyzer or removable media will be damaged.
- Do not place magnetic objects near the monitor. Otherwise, these items may be damaged.

Precautions Against Electromagnetic Interference

- Do not place the analyzer near the source of electromagnetic interference, otherwise it will be interfered with and affect the measurement results, or even cause the analyzer to malfunction.

Important Safety Information

1 . Warning Signs

Please read the following warnings and tips carefully before operating the Automa Hematology Analyzer.
The warning sign on the analyzer or the hazard warning in the user manual consists of a warning sign or a warning sign and a single word （danger, warning, precautions）.



A safety alert symbol. The symbol is located before the word in the hazard warning or appears in the safety-related description in this manual. Please note all relevant safety information after this symbol.



Danger
Indicates an imminently hazardous situation that, if not avoided, will result in death or serious injury. It is not used in this manual.



Warning
Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.



Precautions
Indicates a potentially dangerous situation which, if not avoided, will result in loss of property or data, or cause pollution to the environment.

The following words are used in this manual to indicate prompts:

Precautions:
Represents instructions for use to ensure the correct use of the product to achieve accurate measurement and avoid any failures.

2 . Biological Risk



- Samples, calibrators, controls, waste, etc. are potentially biologically infectious. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment (such as protective clothing and gloves).
- Do not touch the patient's blood specimen directly.
- The surfaces of all parts of the analyzer are potentially biologically infectious, and safety precautions should be taken during operation and maintenance.
- If the analyzer has leaked, the leaked liquid is potentially biologically. infectious.

3. Warning



- This system is limited to professional inspectors, doctors or laboratory operators trained by the Company or an authorized agent.
- If the hospital or institution responsible for this analyzer cannot provide a good maintenance plan, it may cause the normal analyzer to fail and may endanger human health.
- Be sure to use the analyzer under the conditions of use specified in the manual. If the use conditions exceed those specified in the manual, the analyzer may fail to operate normally, the measurement results may be unreliable, and the analyzer components may be damaged and personal injury may occur.
- The analyzer must be used in a well-grounded condition.
- The analyzer must use fuses of the specified specifications.
- Confirm that the voltage meets the requirements.

- Do not place the analyzer near the source of electromagnetic interference, otherwise it will be interfered with and affect the measurement results, or even cause the analyzer to malfunction.
- Do not use the analyzer in a flammable or explosive environment to avoid danger.
- Do not move the analyzer without permission. If you need to move it, please contact our service department.
- The reagent can irritate the eyes, skin, and mucous membranes. Operators should observe laboratory safety regulations when using or contacting items related to reagents, and wear personal protective equipment such as laboratory protective clothing, gloves, etc.
- As soon as the reagent comes into contact with the skin, rinse immediately with plenty of water and seek medical attention if necessary.
- As soon as the reagent comes into contact with the eyes, rinse immediately with plenty of water and immediately get medical attention.
- Dispose of the reagent, waste liquid and waste sample in accordance with regional and national regulations on the discharge and processing of materials with microbial infectivity such as reagent and waste.
- Because the sample probe is thin, and contaminated by calibrators, controls, and samples potentially biologically infectious, please pay attention during operation.
- Dispose of the packaging of discarded calibrators and controls in accordance with regional and national regulations on the discharge and processing of materials with microbial infectivity such as reagent and waste.
- Only the engineers and operators approved by the Company can perform the operation and maintenance, otherwise harmful radiation emission may be caused.
- It is strictly forbidden to open the laser protective cover during normal work. If you find that the protective cover is loose or has a large gap, please contact our Company

The warnings in this manual are listed below. The protective equipment mentioned below includes medical gloves, goggles, protective suits, and masks.

⚠ Infection Caused by Waste Liquid

- Contact with waste liquid may cause infection.
- Wear protective equipment before performing maintenance and inspections.
- If the waste liquid comes in contact with your body, immediately wash it under running water and disinfect the infected site. If necessary, please consult a doctor for examination.

⚠ Infection Caused by Sample

- Contact with sample may cause infection.
- Wear protective equipment when handling samples.
- If the sample comes in contact with your body, immediately wash it under running water and disinfect the infected site. If necessary, please consult a doctor for examination.

- If the sample sticks to the appliance, wear protective equipment and wipe off the solution immediately to disinfect the contaminated part.

⚠ Infections/Injuries from Contact with the Stations of the Analyzer

- Contact with the sample station, reagent filling station, cleaning station, or other stations may cause infection or injury.
- When performing maintenance and inspections, be sure to wear protective equipment and follow the procedures specified in this manual.
- Before starting maintenance, make sure the analyzer is in standby.
- Contact with the sample rack (also called tube rack) and sample rack tray may cause injury or infection.
- When the analyzer is running, switch to emergency mode to put or remove the samples. Before putting or removing samples, withdraw the sample rack. (also called tube rack) to a safe place in the sample rack tray.
- If an alarm occurs on the sample arm and the sample rack, the sample cannot be put in or taken out even in the emergency mode.

⚠ Skin Damage Caused by Acidic or Alkaline Chemicals

- Direct contact with diluent or cleaning solution may cause skin damage.
- When handling diluent or cleaning solution, make sure to wear protective equipment and follow the precautions marked on the label of the vial.
- If the diluent or cleaning solution comes in contact with your body, immediately wash it under running water and disinfect the infected site. If necessary, please consult a doctor for examination.
- When handling reagents, be sure to wear protective equipment and follow the precautions listed in the reagent manual.

⚠ Electric Shock Caused by High Voltage

- Because some components in the power module are under high voltage, there is a danger of electric shock.
- Never remove the cover of the power module.

⚠ Burns Caused by Burning

- Dye solution is a flammable substance, which can easily cause fire or burns if not handled properly.
- When using dye solution for cell staining, do not approach the source of ignition.
- The dye solution is used for cell staining, and the user should be responsible for checking and processing.

4.Precaution

Precautions

- This manual is intended for the following laboratory professionals: Personnel who perform daily system operations

  Personnel who perform system maintenance and troubleshooting

Personnel who learn system operation.

- Please follow the manual strictly during operation.
- This analyzer is for clinical screening. When doctors make interpretations based on the analysis results, they are required to consider the results of clinical examination or other tests.
- Please refer to the reagent instructions for storage and use of reagents.
- Check the balance of the reagent before using the analyzer every day and estimate whether it is sufficient for the use of the specimens of the day. If there is insufficient, you should make preparations in time.
- After replacing any reagent, a blank count test should be performed to ensure that the measured value is within the allowable range of the blank count, and the specimen test can be performed after that.
- The reagents can be used only after recovered from storage temperature to room temperature.
- Unpacking or implementing the installation process by personnel not authorized or trained by the Company may cause damage to the analyzer. Please do not unpack or install the analyzer without the authorized personnel of the Company.
- Do not place reagents or other liquids on top of the analyzer.
- The operator should use clean EDTA-K2 anticoagulation vacuum blood collection tube, siliconized glass tube, plastic tube, centrifuge tube and borosilicate glass capillary.
- Blood collection volume is not less than 1mL in manual whole blood mode.
- Samples requiring leukocyte classification or platelet count should be stored at room temperature and tested within 4 hours of collection.
- If platelet, average red blood cell volume or white blood cell classification results are not needed, the samples can be stored at 2 C to 8 C for 24 hours. Refrigerated samples should be placed at room temperature for at least 30 minutes before testing.
- After the sample is allowed to stand for a period of time, it should be remixed before testing.
- After the peripheral blood is mixed with the diluent, it should be left for 3 minutes for testing.
- Ensure that peripheral blood samples are tested within 30 minutes of dilution.
- The laboratory should evaluate the stability of the test results in the pre-dilution mode according to the number of samples, the method of sample collection, and the technical level.
- There is no sequence for software startup and analyzer startup, and either of them can be started first.
- After a sample is analyzed, the status bar shows that the specimen number will automatically increase by +1. Therefore, as long as the work sheet is registered in order, it is unnecessary to open the manual specimen number for intervention during the test.
- To ensure reliable and stable performance of the analyzer, be sure to perform the shutdown procedure.

such as adding work sheets, viewing results, and printing results without interaction with the analyzer.

- The specimen numbers starting with "QC-" are the quality control data, and you can modify them. You can choose to print in the Result Details or History Data, or you can output in the quality control print format on the QC chart interface.
- The validated status is indicated by a "V" in the validated column.
- Before performing a report printout, confirm the printer installation, connection status, and adequacy of paper supplies.
- The file name is automatically generated by the software, and the user can also rename it as needed.
- Click the graph to enlarge it.
- Only the user with administrator rights can perform calibration operations.
- The user should use the control and calibrator specified by the Company and store them strictly in the storage environment specified by the instructions for calibrator, control and reagent.
- Use only the specified controls. Controls are designed to monitor the status of the analyzer.
- When the batch number is not entered, the "OK" button is disabled.
- You can modify the registered document number by clicking "Modify". The quality control materials cannot be modified.
- The figure "1" in the confirmation dialog box indicates the number of documents currently selected for deletion.
- Improper maintenance may cause damage to the analyzer. The operator must perform maintenance in accordance with the manual.
- If you encounter a problem that is not clear in the manual, please contact our service department, and the Company's designated professional will give maintenance suggestions.
- The analyzer must be maintained using parts provided by the Company. If you have any questions, please contact our service department.
- This manual is not a maintenance manual, and only provides the measures.

The notes in this manual are listed below. The protective equipment mentioned below includes medical gloves, goggles, protective suits, and masks.

⚠ Pollution due to Improper Disposing of Waste

- Waste liquid containing infectious pathogens blood samples must be disposed of in accordance with relevant laws and regulations under the guidance of personnel responsible for the management of infectious medical waste
- Substances contained in reagents, calibrators, and controls are subject to environmental regulations, and must be disposed of in accordance with the wastewater quality standards of local environmental protection agencies.

⚠ Improper Disposing of Infectious Waste

- Waste materials are infectious waste and must be disposed of in accordance

with relevant laws and regulations and the safety control procedures of local agencies under the guidance of personnel responsible for the management of infectious medical waste.

Inaccurate Measurement Due to Insufficient Sample Aspiration

- If the sample contains insoluble substances such as fibrin or dust, the measurement may be inaccurate due to insufficient sample aspiration. When placing samples, make sure that they do not contain insoluble substances.
- If air bubbles are formed in the sample container, the measurement may be inaccurate due to insufficient sample aspiration. When placing samples, make sure that they do not contain air bubbles.

Inaccurate Measurement Due to Improper Sample Loading

- If air bubbles are formed in a reagent vial, the measurement may be inaccurate due to insufficient sample aspiration.
- When placing the reagent vial, make sure that it does not contain air bubbles.

Sample Misreading Caused by Manual Registration

- If the entry position is wrong when the sample is manually registered, the measurement result of other sample is obtained.
- Place the sample in the sample rack also called tube rack according to the procedure specified in this manual, and confirm that the placed sample matches the set information.

Reagent Misreading Caused by Manual Registration

- If the barcode is entered incorrectly when manual registration is applied for reagent, the measurement will be incorrect due to the reagent used that does not match.
- When manually registering the reagent, please enter the barcode on the reagent barcode label correctly according to the procedure specified in this manual.

Calibrate or Control Misreading Caused by Manual Registration

- If the information is entered incorrectly when manual registration is applied for calibrator or control, the measurement will be incorrect due to the calibrator or control used that does not match.
- When manually registering the calibrator or control, please enter the information of the calibrator or control according to the procedure specified in this manual.

Fatigue from Prolonged Use

- Staring at the LCD monitor for a long time with the same gesture would increase body and eye fatigue. Therefore, please take a rest for 10 to 15 minutes after every one hour of continuous working in front of the LCD monitor. Do not staring at the LCD monitor for over six hours daily .

Chapter 1 Overview of the Manual

1.1 Overview

This chapter explains how to use the user manual of the Automatic Hematology Analyzer. This manual, delivered with the analyzer, explains the purpose, function, and usage of the analyzer. Before using the analyzer, please read and understand this manual carefully to ensure that you can use the analyzer correctly, exert its best performance, and ensure the safety of the operator.

Precautions

- Please follow the manual strictly during operation.

1.2 Contents of the Manual

This manual contains 10 chapters and 1 appendix. Find the appropriate chapters as needed.

Chapter	Contents of Chapters
Chapter 1 Overview of the Manual	Outlines the manual of the analyzer
Chapter 2 System Introduction	Explains the purpose and measurement parameters of the fully automated analyzer Explains the structure and operation interfaces of the analyzer
Chapter 3 Design Principles	Explains the measuring principle and workflow of the analyzer
Chapter 4 System Installation	Describes the installation requirements and methods of the analyzer
Chapter 5 Basic Settings	Explains system settings such as the range of reference values in the analyzer
Chapter 6 Daily Operations	Explains the sample analysis process and daily operation
Chapter 7 Quality Control	Explains the quality control requirements and methods of the analyzer
Chapter 8 Calibration of the Analyzer	Describes the calibration requirements and methods of the analyzer
Chapter 9 User Services	Explains how to maintain, service and test the analyzer
Chapter 10 Troubleshooting	Explains the cause of the failure and the troubleshooting method for the analyzer
Appendix	Describes the specifications of the analyzer

1.3 Scope of Application

This manual is intended for medical laboratory professionals or trained doctors, nurses, and laboratory technicians. It is used for:

- Understanding the hardware and software of the analyzer
- Setting the system parameters
- Daily analysis of samples
- System maintenance and troubleshooting.

1.4 Symbol Description

1.4.1 Symbols in the Manual

The following symbols are used in this manual, and their meanings are explained as follows:

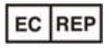
Symbol	Meaning
⚠ Warning	Prompts the operator to follow the instructions under the symbol, otherwise personal injury may be caused.
⚠ Caution	Prompts the operator to follow the instructions under the symbol, otherwise it may cause failure, damage to product or affect the test results.
Precautions	Prompts the operator to follow the instructions under the symbol, emphasizing important information in the operation steps or content that requires the operator's special attention.
	Prompts the operator to follow the instructions under the symbol, otherwise there is a danger of potential biological infectivity.
	Never open the door or cover during normal work. If the door or cover needs to be opened, it must be carried out by personnel who have received relevant training and approval after the analyzer is properly shut down.

1.4.2 Analyzer System Symbols

1.4.2.1 Symbol

The following symbols may be seen on the analyzer system, reagent kit, control or calibrator and their packages:

Symbol	Meaning
	Warning Symbol Neighboring areas of this label may lead to death or serious injury. Please refer to the manual to make sure correct use of the product.
	Biohazard Warning Neighboring areas of this label have potential biohazard. Relevant lab safety procedures must be followed.
	High Voltage Warning Neighboring areas of this label have high-voltage parts. Contact may cause electric shock.
	Laser Warning Neighboring areas of this label have laser parts. Contact may cause laser damage.
	Mechanical part Contact with any moving mechanical parts in neighboring areas of this label may be dangerous.
	Away from water Penetration of liquids into neighboring areas of this label may damage the analyzer. Keep the area away from any liquid.
	High Temperature Warning
	Protective Grounding
	AC
	Manufacturer

	Refer to the user manual
	Serial Number
	In vitro diagnostic use only
	Storage temperature
	Batch No.
	Anti-static label
	Catalog number
	Authorized representative in the European Community
	Do not discard it carelessly, hand it over to appropriate facilities

1.4.2.2 Laser

Laser Type	Wavelength	Power	Maximum Output Power	Horizontal Transmitting Angle	Vertical Transmitting Angle
Class 3B laser	635nm	3.5mW	5mW	8°	31°

You may see the following laser warning labels on the outside or inside the analyzer. Please follow the instructions to prevent laser hazards!



This label is posted on the laser shielding box of optical components



This analyzer is a Class 1 laser product, and this label is affixed to the body enclosure



This label is affixed to the body enclosure

Laser Spot Size:



1.4.2.3 Nameplate

The nameplate on the back panel of the analyzer indicates the analyzer model and SN.

1.4.2.4 Symbol Position

The labels affixed to the Automatic Hematology Analyzer are listed below. Read each label carefully and confirm the contents on the actual parts.

Check the labels frequently, and clean and maintain them so that they can be clearly read from a safe distance. If the labels are not easily recognizable due to aging, please contact our service department to replace them with new ones.

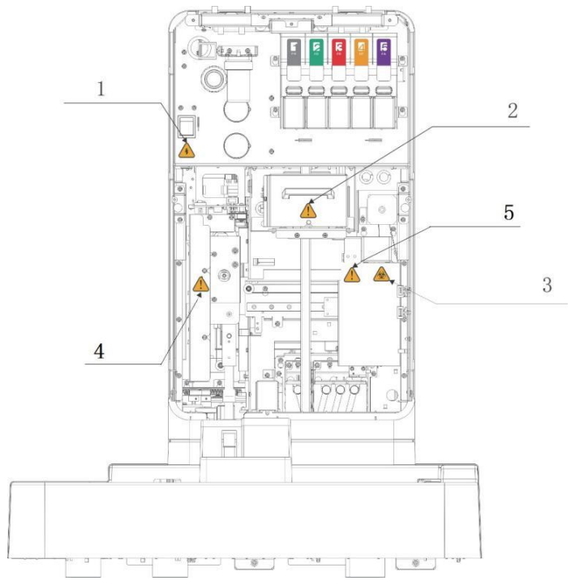


Figure 1-1 Front view of the analyzer （ open the upper cover of the front shell ）

1



There is a danger of electric shock in operation of the power switch.

2,4,5



To avoid damage to the analyzer, do not disassemble this part.

3



There is a danger of potential biological contagion, please read the manual carefully.

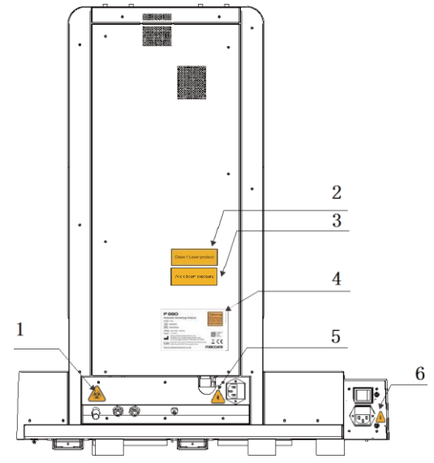


Figure 1-2 Back view of the analyzer

1



There is a danger of potential biological contagion, please read the manual carefully.

2



This analyzer is a Class 1 laser product.

Template of the analyzer.



There is a danger of electric shock in operation of the power switches of the main unit and the sample loader.

Figure 1-3 Right side view of the analyzer （ open the cover ）

1



Note: There is laser radiation when opening, and avoid beam

2



Laser warning label, please follow the manual to prevent laser hazards.

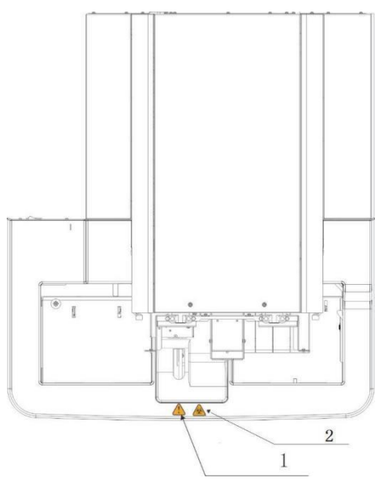


Figure 1-4 Sample loader

1



There is a mechanical danger in removal of this protective cover.

2



There is a danger of potential biological contagion, please read the manual carefully.

Chapter2 System Introduction

2.1 Overview

The Automatic Hematology Analyzer is an in vitro diagnostic device. It is used with supporting reagents, calibrators, and controls for the quantitative analysis of blood cells in a medical laboratory.

According to the differences in channels, quality control methods and the number of sample racks also called tube racks, the analyzer is divided into 6 models:

Measurement channels and detection baths				Model specification		
Channel	Abbreviation	Detection bath	Abbreviation	HX 88	HX 85	HX 80
Complete blood count channel	CBC	White blood cell count detection bath including RBC and BASO counts	FCW	√		
		Red blood cell/platelet detection bath	RBC/PLT	√		
		Hemoglobin concentration detection bath	HGB	√		
Differential white blood cell WBC count channel	DIFF	Differential WBC detection bath	FCD	√		
Abnormal leukocyte primary screening channel	AWS	Abnormal cell detection bath	FCA	√	–	–
Reticulocyte count channel	RET	Reticulocyte detection bath	FCR	√	√	–
Fluorescent platelet confirmation channel	PCF	Fluorescent platelet detection bath	FCP	√	–	–
QC methodl	L-J			√		
	X-bar			√	√	√
	X-barM			√		
umber of sample racks also called tube racks				5	5	5

2.2 Scope of Application

The product is suitable for quantitative or qualitative analysis of the constituents in human blood samples and provides relevant information.

Precaution

- This analyzer is for clinical screening. When doctors make interpretations based on the analysis results, they are required to consider the results of clinical examination or other tests.
- The analyzer is used for clinical anticoagulation detection for the samples without coagulation or hemolysis.

The analyzer provides:

- 37 reportable parameters
- 34 research parameters
- 2 histograms
- 5 scattergrams
- 12 detection modes
- 4 sample modes

Reportable Parameters

Item		Model specification		
English Name	Short Name	HX88	HX 85	HX 80
White blood cell count	WBC	√	√	√
Number of basophils	BASO#	√	√	√
Percentage of basophils	BASO%	√	√	√
Number of neutrophils	NEUT#	√	√	√
Percentage of neutrophils	NEUT%	√	√	√
Number of eosinophils	EO#	√	√	√
Percentage of EO	EO%	√	√	√
Number of lymphocytes	LYMPH#	√	√	√
Percentage of lymphocytes	LYMPH%	√	√	√
Number of monocytes	MONO#	√	√	√
Percentage of monocytes	MONO%	√	√	√
Number of immature granulocytes	IG#	√	√	√
Percentage of granulocytes	IG%	√	√	√
Number of red blood cells	RBC	√	√	√
Hemoglobin concentration	HGB	√	√	√
Mean corpuscular volume	MC √	√	√	√
Mean corpuscular hemoglobin content	MCH	√	√	√
Mean corpuscular hemoglobin concentration	MCHC	√	√	√
Red blood cell distribution width coefficient of variation	RDW-C √	√	√	√
Red blood cell distribution width standard deviation	RDW-SD	√	√	√
Hematocrit	HCT	√	√	√
Number of nucleated red blood cells	NRBC#	√	√	√
Percentage of nucleated red blood cells	NRBC%	√	√	√
Platelets	PLT	√	√	√
Mean platelet volume	MP √	√	√	√
Platelet distribution width	PDW	√	√	√
Platelet hematocrit	PCT	√	√	√
Platelet-large cell ratio	P-LCR	√	√	√
Platelet-large cell count	P-LCC	√	√	√

Immature platelet fraction	IPF	√	-	-
Reticulocyte percentage	RET%	√	√	-
Number of reticulocytes	RET#	√	√	-
Immature reticulacyte fraction	IRF	√	√	-
Low fluorescent reticulocyte ratio	LFR	√	√	-
Middle fluorescent reticulocyte ratio	MFR	√	√	-
High fluorescent reticulocyte ratio	HFR	√	√	-
Reticulocyte hemoglobin concentration	RET-He	√	√	-

Research

Item	Model specification			
Parameter (English /ame)	Parameter	HX 88	HX85	HX 80
Number of platelets calculated from RBC/PLT channel	PLT-I	√	√	√
Number of platelets calculated from FCR channel	PLT-O	√	√	-
Number of platelets calculated from FCP channel	PLT-F	√	-	-
Number of WBC calculated from FCW channel	WBC-W	√	√	√
Number of WBC calculated from FCD channel	WBC-D	√	√	√
Total number of nucleated cells	TNC	√	√	√

T C calculated from FCW channel	TNC-W	√	-
Total of nucleated cells calculated from the FCD channel WBC + RBC	TNC-D	√	-
Number of RBC calculated from FCR channel	RBC-O	√	-
Micro red blood cell ratio	microR	√	-
Macro red blood cell ratio	macroR	√	-
Number of granulocytes obtained by subtracting the number of IGs from the number of E Ts	NEUT#&	√	-
Ratio of E T number minus IG number to WBC count	NEUT%&	√	-
Number of lymphocytes obtained by subtracting HFLC number from LYMPH number	LYMP#&	√	-
Ratio of LYMPH number minus HFLC number to WBC count	LYMP%&	√	-
Area count above LYMPH in the FCD scattergram	HFLC #	√	-
Ratio of area count above LYMPH in the FCD scattergram to WBC count	HFLC %	√	-
umber of basophils calculated from FCW channel	BA-W #	√	-

Basophil percentage calculated from FCW channel	BA-W%	✓	✓	✓
Number of basophils calculated from FCD channel	BA-D#	✓	✓	✓
Basophil percentage calculated from FCD channel	BA-D%	✓	✓	✓
Correlation between RBC-Y forward scattered light intensity of mature red blood cells and MCH to convert RBC-Y to [pg] unit	RBC-He	✓	✓	–
Calculated by the equation Delta-He = RET-He-RBC -He	Delta-He	✓	✓	–
Reticulocyte production index	RPI	✓	✓	–
Ratio of the number of platelets in the area of strong fluorescence intensity to the total number of platelets in the IPF region on the FCP scattergram	H-IPF	✓	–	–
Number of platelets in the IPF region of the FCP scattergram	IPF#	✓	–	–
Number of WBC calculated from FCA channel	WBC-A	✓	–	–

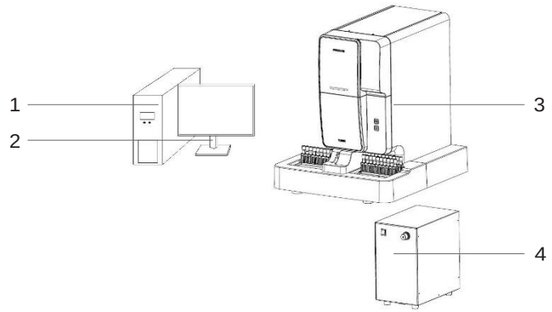
TNC calculated from FCA channel	T C-A	✓	–
Ratio of particle number in the low-value area of the forward scattered light signal in the RBC mature red blood cell region of the FCR channel to RBC	HYPO-He	✓	✓
Ratio of the number of particles in the high value are of the forward scattered light signal in the RBC mature red blood cell region of the FCR channel to RBC	HYPER-He	✓	✓
Count in the UPP region of the FCR scattergram	FCR-UPP	✓	✓
Count in the TNC region of the FCR scattergram	FCR-TNC	✓	✓
Absolute number calculated from the number of particles in a specific area below the RBC particles on the FCR scattergram	FRC#	✓	✓
Percent calculated from the number of particles in a specific area below the RBC particles on the FCR scattergram	FRC%	✓	✓

2.3 Product Structure

2.3.1 Overview

The product is mainly composed of analysis module, sample loader(optional) , compressor unit, and analyzer software release version number is V1 . The analysis module includes a fluidic module, an optical module, a sampling module, a mixing module, and a control module.

The main components of the detection system are as follows:



1, 2 Analyzer Control System DCS 3 Analysis Module and Sample Loader 4 Compressor Unit

- Analyzer Control System(DCS)

  Controls the analysis module and the sample loader, processes and displays the original measurement, edits the worksheet, views the history and detailed sample results, prints the sample result report, etc.

- Analysis Module and Sample Loader

  Processes and analyzes samples and sends raw measurement data to DCS.

- Compressor nit

  Provides positive and negative pressure to the analysis module.

2.3.2 Major Modules and Components

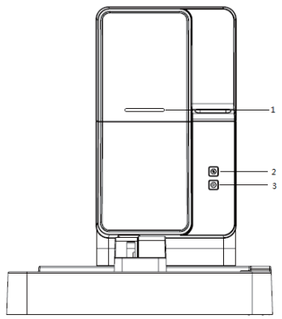


Figure 2.3-1 Front view

1 Status indicator 2 Mode switch button 3 Start measurement button

- Status indicator

  The status indicator is located on the front of the analyzer and shows whether the analyzer is currently ready or not ready.

- Mode switch button

  Switch between auto and manual loading mode.

- Start measurement button

  The start measurement button is located on the front of the analyzer and triggers the analyzer to start the corresponding sequence.

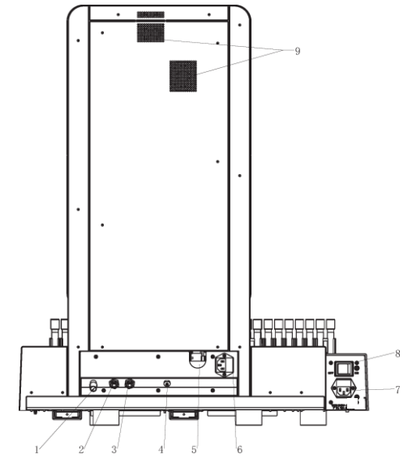


Figure 2.3-2 Rear view

1 Waste liquid port	2 Positive pressure port	3 Negative pressure port
4 Diluent port	5 Line outlet	6 Power supply port of analysis module
7 Power supply port of sample loader	8 Power switch of sample loader	9 Fan vent for analysis module

- Waste liquid port

  Connects with waste liquid tube to discharge waste liquid from the analyzer.

- Positive pressure port

  Connects with the positive pressure port of the compressor unit to provide positive pressure to the analyzer.

- Negative pressure port

  Connects with the negative pressure port of the compressor unit to provide negative pressure to the analyzer.

- Diluent port

  Connects with an external dilution container to provide the analyzer with the required diluent.

- Line outlet

  Connects with the reagents outside the analyzer, and provides the required reagents and the data connection line for the analyzer.

- Power supply port of analysis module

Inserted with power cable that is connected with the power supply.

- Power supply port of sample loader

  Inserted with power cable that is connected with the power supply

- Power switch of sample loader

  Controls the power supply of sample loader.

- Fan vent for analysis module

  Exhausts hot air inside the analyzer.

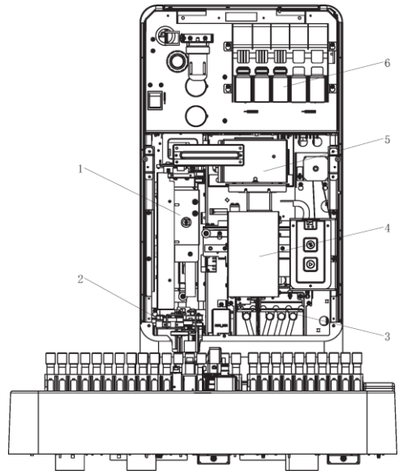


Figure 2.3-3 Front view (Remove the front cover)

1 Clamping and mixing module	2 Blood sample collection position	3 DIFF reaction bath
4 Sampling assembly	5 RBC module	6 Reagent storage module

- Clamping and mixing module

  Moves forward and backward / up and down and swings, to transfer the blood collection tube and mix the blood sample in blood collection tube.

- Blood sample collection position

  Place the blood collection tube to facilitate the sampling of sampling assembly.

- DIFF reaction bath

  For the reaction of blood samples and reagents, in preparation for subsequent testing.

- Sampling assembly

  Move left and right and up and down for blood sample collection and distribution.

- RBC module

  Detect the RBC impedance of the analyzer.

- Reagent storage module

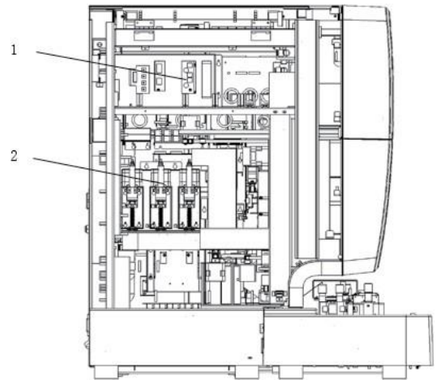


Figure 2.3-4 Left side view (Remove the left cover)

1 Power board 2

- Power board

  Power distribution unit of the analyzer.

- Syringe

  Provides high-precision / stable sample flow for related parts.

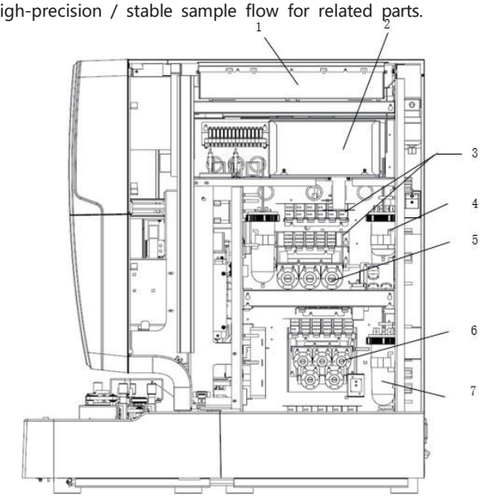


Figure 2.3-5 Right side view Remove the right cover

1 Control box	2 Optical system	3 Solenoid valve
4 Reservoir	5 Dosing pump 1	6 Dosing pump 2
7 Waste container		

- Control box

  Place the control board and driver board of the analyzer.

- Optical system

Laser component that uses a laser to detect the sample.

- Solenoid valve

  Control the on-off of the tubes to achieve the established fluidic control.

- Reservoir

  Stores the liquid to provide a stable liquid to the analyzer.

- Dosing pump 1

  Provides high-precision and quantitative samples for the analyzer.

- Dosing pump 2

  Provides high-precision and quantitative samples for the analyzer.

- Waste container

  Stores the waste liquid generated by the analyzer and connects with the waste liquid channel outside the analyzer to discharge the waste liquid from the analyzer.

2.4 Operation Interface

Main Interface

Double-click the desktop icon to enter the main interface after the startup process is completed. By clicking an icon on this interface, you can access the corresponding function interface.

Level	Level	Function
Patient list	Patient info.	Registers or modifies patient information
	Bed o.	Registers or modifies bed o.
	Dr. name	Registers or modifies physician name
Worksheet		Registers or modifies worksheet sample
Result details	Comprehensive	Views the results, IP alarms, actions, and error / rule descriptions of a measured sample in detail Reviews, prints, and outputs samples
	Graph	View the result, IP alarm, histogram and scattergram of a measured sample in detail Reviews, prints, and outputs samples
	Comparison	Views the results of multiple measurements of a patient, trend graphs and scattergrams in detail
	Alarm value	Checks the positive levels of 11 suspicious IPs of a measured sample in detail
	Maintenance	Views the related maintenance data of a measured sample in detail
	Custom parameter	For laboratory personnel to customize the layout of all parameters that have the analyzed results
History	Sample information	Views the sample information of multiple samples Reviews, prints, and outputs samples
	CBC	Views CBC results of multiple samples
	DIFF	Views DIFF results of multiple samples
	RET*	Views RET results of multiple samples
	PCF*	Views PCF results of multiple samples
	Patient info.	Views the patient information of multiple samples
	Reagent	Views the reagent information of multiple samples
Precision		Views the measurement result of output precision displayed
Calibration		Views the measurement result of output calibration displayed
QC		Registers or views QC document Views QC status with QC chart
System settings		Sets related DCS parameters
Instrument settings		Sets related instrument parameters
Template settings		User can customize the template for printing report forms as required
Log	Operation log	Saves and displays the operation logs performed by users
	Error log	Saves and displays the alarm log of the analyzer
	Reagent log	Saves and displays the log when the user replaces the reagent
	Maintenance log	Saves and displays the maintenance logs performed by users
Retest rule		Views and sets different retest rules
Version information		Displays the specific version number of the software
User manual		Views help tips
Logout		Exits the software and changes the login user
Exit		Closes the DCS software

***This content is not shown on all analyzer models**

Function Interface

- Shortcut button

Name	Function
Menu	Enters the main interface
QC	Registers or views QC document Views QC status with QC chart
Worksheet	Registers or modifies worksheet sample
History data	Views the results of multiple measured samples Reviews, prints, and outputs samples
Result details	Views the result of one measured sample in detail Reviews, prints, and outputs samples

- Analyzer Control Bar

  Controls the analyzer for sample measurement, calibration measurement, control measurement, maintenance, troubleshooting, reagent replacement and shutdown. Displays the analyzer ID, sample information, and analyzer sample loader status.

- Content display area

  Displays the corresponding function.

- Toolbar

  Displays the function buttons available in the current interface.

- Information status bar

  Displays login user, printer, main unit status, and displays company, page and date information.

2.5 General Operations

Name	Operation Performed
Click	Click on the corresponding location or button, and sometimes double-click the button.
Input	Click the input or edit box to make the cursor appear in the input box and use the keyboard to enter the required information letters, characters, numbers, symbols, etc. .
Cancel	Click (or use the ↑, ↓,←and →keys on the keyboard) to position the cursor where you want to delete. The Backspace key is used to delete the information before the cursor, and the Delete key is used to delete the information after the cursor.
Drag the scroll bar	Click the arrows on both sides of the scroll bar to perform the page turning operation; Or click and select the scroll bar and move the mouse to turn the page.
Select the drop-down list	Click the down arrow of the drop-down box. The drop-down list is popped up. Click to select or use the ↑ and ↓ keys on the keyboard to select or touch to move by finger and select.

2.6 Sample Keyword Description

2.6.1 Description of Sample No.

Samples can be loaded by two ways, autoloading or manual loading. Control samples, calibration samples, and blank count detection samples are all classified as a manual loading type.

The sample No. can be obtained by auto increment, scanning, manual input, and main unit query. Scanning is performed by the barcode scanning module that comes standard with the analyzer.

The worksheet sample is registered before the measurement or before the result is obtained during the measurement. The untested worksheet sample No. should not be repeated.

The sample No. starts from "1" every day. If the sample "1" has been registered on the worksheet, the measurement starts from the worksheet. The auto- increasing sample is automatically incremented during the measurement (the user can set the No. from which to start the measurement before the measurement), and the next sample is plus 1 on the previous one. For example, if the previous one is "s1", the latter is "s2".

The sample No. generated by scanning is the barcode number, which is generated by the analyzer during the measurement process. The scanning matching method needs the support of the scanning module of the analyzer.

2.6.2 Description of Position No. in the Sample Rack (Also Called Tube Rack)

1. The generation method of the sample rack No. and position No. is divided into 2 types. The sample rack No. and position No. can be edited when the user registers the worksheet sample or can be automatically incremented during the measurement. The sample rack No. and position No. of the worksheet sample that has not been measured should not be repeated.
2. When the automatically generated sample rack No. is generated, if the position No. of the previous sample rack is 10, 1 is added to the previous one. If the position No. of the previous sample rack is not 10, the previous sample rack o. is maintained. The sample rack o. starts from 1 each day.
3. When the automatically generated position No. of the sample rack is generated, if the position No. of the previous sample rack is 10, the No. starts from 1. If the position No. of the previous sample rack is not 10, 1 is added to the previous one. The position No. of sample rack starts from 1 each day.

2.7 Reagent, Calibrator and Control

The analyzer, reagent, calibrator, and control together form a system and must be used as a whole to ensure overall performance.

The reagents specified by the Company must be used, otherwise the reliability and accuracy of the analysis results cannot be guaranteed, and the analyzer may be damaged. "Reagents" appearing in this manual refer to the reagents used with this analyzer.

Each reagent must be checked for package integrity and expiration date before use. Make sure that the package is not wet or leaking and that the reagents

Precautions

- Please refer to the reagent instructions for storage and use of reagents.
- Check the balance of the reagent before using the analyzer every day, and estimate whether it is sufficient for the use of the specimens of the day. If there is insufficient, you should make preparations in time.
- After replacing any reagent, a blank count test should be performed to ensure that the detection value is within the allowable range of the blank count, and the specimen test can be performed after that.
- The reagents should be used only after recovered from storage temperature to room temperature.

2.7.1 Reagent

- Automatic Hematology Analyzer DK Diluent

  The diluent is an isotonic liquid with a specific electrical conductivity, providing a stable environment for blood cell counts, and performing fludics cleaning after the test is complete.
- Automatic Hematology Analyzer DR Diluent
  DR is used in combination with FR to measure reticulocytes, or in combination with FP to measure platelets.
- Automatic Hematology Analyzer LS Lysing Reagent Lyses red blood cells and measures hemoglobin.
- Automatic Hematology Analyzer LW Lysing Reagent

  Lyses red blood cells and counts white blood cells and nucleated red blood cells.
- Automatic Hematology Analyzer LD Lysing Reagent Lyses red blood cells and differentiates white blood cells.
- Automatic Hematology Analyzer LA Lysing Reagent Lyses red blood cells and differentiates some abnormal cells.
- Automatic Hematology Analyzer FW Fluorescent Dye

  Stains leucocyte nucleic acid, and counts leukocytes and nucleated red blood cells.
- Automatic Hematology Analyzer FD Fluorescent Dye Stains

  Stains nucleic acid on nucleated red blood cells and platelets, and counts nucleated red blood cells and platelets.
- Automatic Hematology Analyzer FP Fluorescent Dye Stains nucleic acid on platelets and counts platelets.
- Automatic Hematology Analyzer FA Fluorescent Dye

  Stains some abnormal cells and differentiates some abnormal cells.
- Automatic Hematology Analyzer CC Cell Clean
  Contains a detergent that removes stains from the pipeline for routine cleaning and maintenance of the analyzer.

2.7.2 Control and Calibrator

Control and calibrator are used for quality control and calibration of the analyzer.

The "control" and "calibrator" mentioned in this manual refer to the supporting control and calibrator, which should be bought from Agappe or its Distributors

Chapter 3
Design Principles

3.1 Overview

The analyzer uses sheath flow impedance method to detect the number of red blood cells and platelets colorimetric method to measure hemoglobin and semiconductor laser flow cytometry to detect white blood cells, nucleated red blood cells and reticulocytes.
And calculates the remaining parameters according to the corresponding formula.

3.2 Sample Aspiration

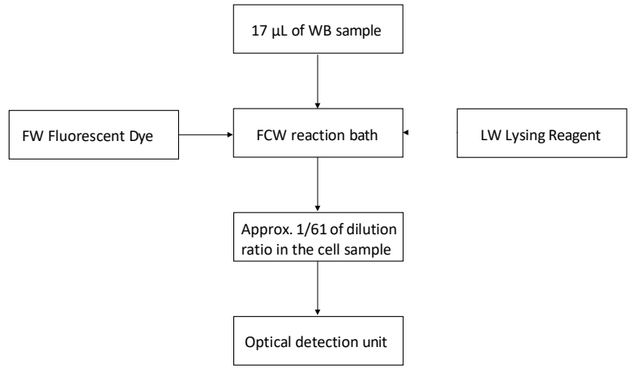
The analyzer provides two modes of manual loading and autoloading. Manual loading is divided into two sub modes: closed loading and open cover loading.
Manual loading supports four modes basing on sample types: whole blood WB sampling mode, low value white blood cell LW sampling mode, pre-dilution (PD) sampling mode and sample research (BF) sampling mode.
Auto-loading supports 2 modes of WB and LW.
In WB mode and LW mode, the analyzer will aspirate 88 μL of WB sample.
In the PD mode, the operator first mixes 20 μL of blood sample with 120 μL of diluent outside the analyzer to form a 1: 7 dilution sample for testing.

3.3 Sample Mode

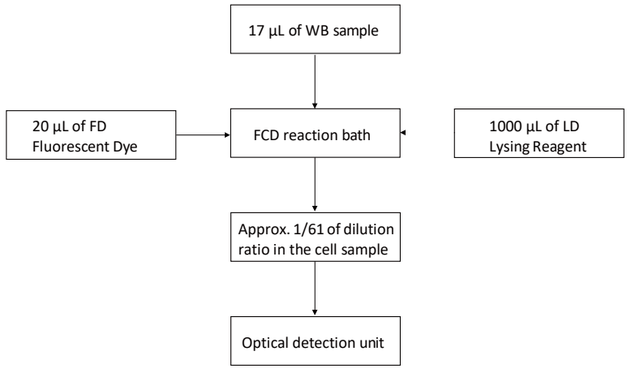
After the blood sample is collected by the analyzer, the blood is distributed and diluted according to the measurement requirements of different parameters, which are respectively used for hemoglobin count, red blood cells count, platelets count, white blood cell count, and white blood cell differentiation measurement.
According to different choices of users, the analyzer provides four working modes basing on sample types: WB working mode, LW working mode, PD working mode, and BF working mode.

3.3. Whole blood mode

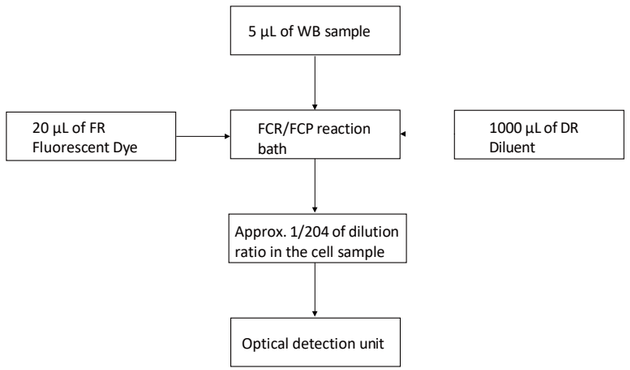
FCW dilution process



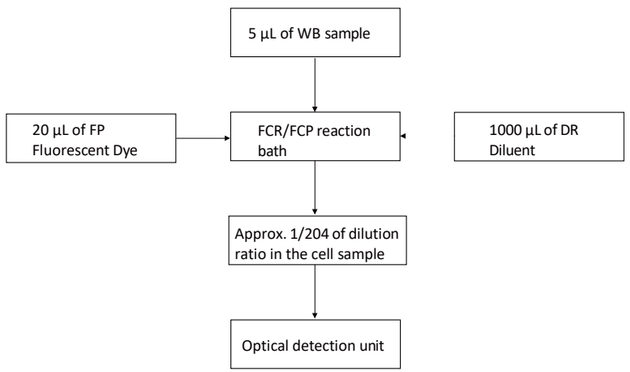
FCD dilution



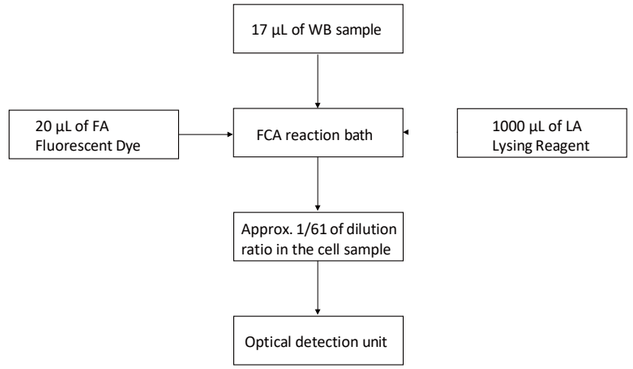
FCR dilution



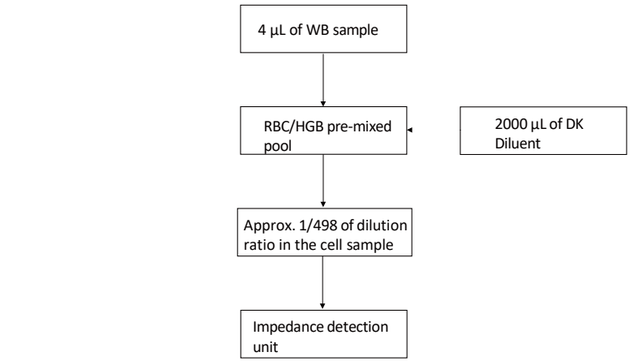
FCP dilution



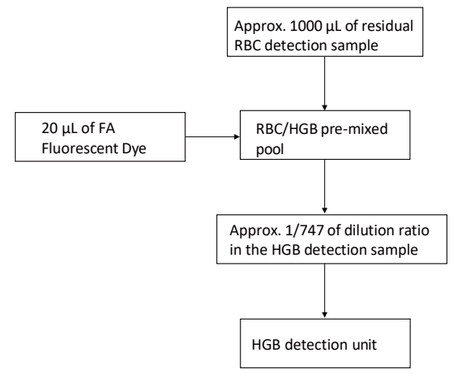
FC dilution



RBC/PLT dilution



HGB dilution process



3.4 Measurement Principle

Laser Flow Cytometry and Fluorescence Staining Detection Technology

Flow cytometry measures the physical and chemical properties of cells and other biological particles. This method requires fewer samples for detection. After a certain amount of blood cells have been aspirated and reacted with a specific reagent, a certain amount of blood specimen is injected through a nozzle into a conical flow chamber filled with diluent. Encapsulated by the sheath fluid formed by the diluent, the cells are arranged individually in rows through the center of the flow chamber, as shown in Figure 3 1 Schematic
diagram of laser flow cytometry.

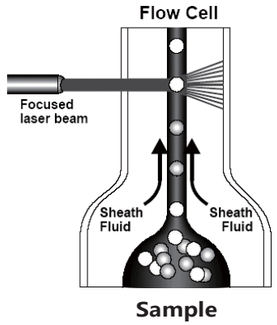


Figure 3-1 Schematic diagram of laser flow

When the blood cells suspended in the sheath fluid pass through the laser detection area, the blood cells are irradiated by the laser beam, and the properties of the scattered light are related to the cell size, the cell membrane and the refractive index of the internal structure of the cell. The photodiode or photomultiplier tube receives these scattered light signals and converts them into electrical pulses. Based on the collected electrical pulse data, a two- dimensional distribution diagram of two of the parameters of blood cell size, complex structure inside the cell, or nucleic acid content can be obtained, which is called a scattergram.

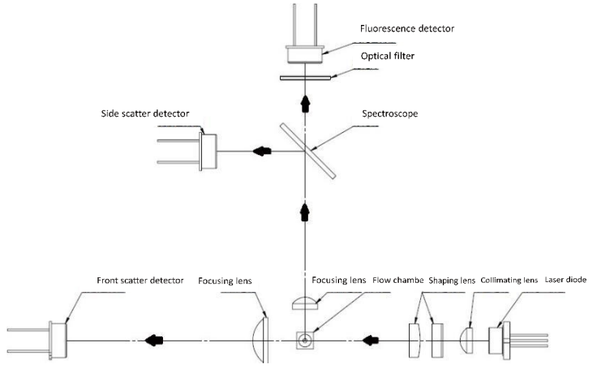


Figure 3-2 Schematic diagram of fluorescent staining technology

The forward scattered light FSC reflects the size of the cell, the side scattered light SSC reflects the internal complex structure and particulate matter of the cell, and the side fluorescence SFL reflects the internal

Sheath Flow Impedance Method

Count RBC / PLT by sheath flow impedance method. Based on the impedance method, the sheath flow impedance method uses the FCM sheath flow technology, which makes the cells encapsulated by sheath fluid to pass from the center of the aperture. Within the detector, the reagent nozzle will be positioned in front of the aperture and aligned with the center. After the diluted sample is pushed into the conical chamber by the reagent nozzle, it is encapsulated by the sheath flow and passes through the center of the aperture.

After passing through the aperture, the cells enter the recovery tube in time, preventing the cells from returning to the current-affected area the aperture and its vicinity. At the same time, the accuracy, stability and repeatability of blood cell counting are improved, and the probability of clogging is reduced.

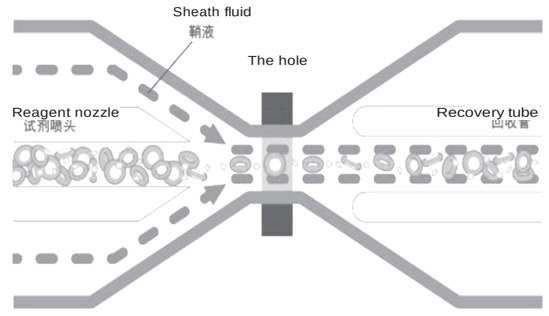


Figure 3-3 Schematic diagram of sheath flow impedance method

Colorimetry

In the hemoglobin HGB colorimetric bath, after the diluted sample is added with a lysing agent, red blood cells are lysed and hemoglobin is released. The latter combines with the lysing agent to form a hemoglobin complex. At one end of the colorimetric bath, the LED light tube irradiates the hemoglobin complex solution after passing through the monochromatic light-emitting tube at a wavelength of 525 nm, and the photoelectric cell at the other end receives the transmitted light, and amplifies the light intensity signal and converts it into a voltage signal. This voltage signal is compared with the voltage generated by the blank transmitted light intensity before the sample is added to the bath there is only diluent in the bath. Then the hemoglobin concentration of the sample is obtained.

3.5 Measurement Items and Channels

3.5. WBC Measurement

■ FCW Channel

This channel divides cells into nucleated red blood cells（NRBC）, basophils, and white blood cells other than basophils.

Application scope of FCW channel: When NRBC# is 0.0 – 20.00 X 10^9/L, the linear correlation coefficient is r ≥ 0.990, the allowable deviation range is ± 20.0% or ± 2.00 X 10^9/L.

When the NRBC% is 0.0 - 600.0/100WBC, the linear correlation coefficient is r ≥ 0.990, and the allowable deviation range is ± 20.0% or ± 2.0NRBC%. The accuracy of NRBC% satisfies the correlation coefficient r ≥ 0.80.

■ **FCD Channel**

Analyzes the scattergram and its neutrophil area, lymphocyte area, monocyte area, eosinophil area, and immature granulocyte area to obtain EEUT%, LYMPH% , MONO% , EO% , and IG% . Combined with the total number of white blood cells output from the FCW channel, calculates NEUT# ,LYMPH# , MONO# , EO# , BASO# , and IG# . The unit of the number of cells is 10^9/L by default.

a)

$$NEUT\% = \frac{\text{Number of particles falling in the E T region in the FCD channel}}{\text{Total number of white blood cells in the FCD channel after corrected by nucleated red blood cells}} \times 100\%$$

b) **Number of neutrophils**

$$NEUT\# = NEUT\% \times WBC$$

c) **LYMPH %**

LYMPH= umber of particles falling in the LYMPH region after corrected by nucleated red blood cells

$$LYMPH\% = \frac{LYMPH}{\text{Total number of white blood cells in the FCD channel after corrected by nucleated red blood cells}} \times 100\%$$

d) **Number of lymphocytes**

$$LYMPH\# = LYMPH\% \times WBC$$

e) **MONO%**

$$MONO\% = \frac{}{\text{Total number of white blood cells in the FCD channel after corrected by nucleated red blood cells}} \times 100\%$$

f) Number of monocytes

$$MONO\# = MONO\% \times WBC$$

g)

$$EO\% = \frac{\text{Number of particles falling in the EO region in the FCD channel}}{\text{Total number of white blood cells in the FCD channel after corrected by nucleated red blood cells}} \times 100\%$$

h) Number of eosinophils

$$EO\# = EO\% \times WBC$$

i) BASO%

$$BASO\% = \frac{\text{Total number of BASO in the FCW channe}}{\text{Total number of white blood cells in the FCW channel}} \times 100\%$$

J) Number of basophils

$$BASO\# = BASO\% \times WBC$$

k) IG%

$$IG = \frac{\text{Number of particles falling in the IG region in the FCD channel}}{\text{Total number of white blood cells in the FCD channel after corrected by nucleated red blood cells}} \times 100\%$$

l) Number of immature granulocytes

$$IG\# = IG\% \times WBC$$

■ **FCA Channel**

A channel for detecting immature cells such as bone marrow bulbs and abnormal cells of the lymphoblastic system.

3.5.2 RBC/PLT Sheath Flow Impedance Measurement

a) **Number of red blood cells**

The analyzer obtains the number of red blood cells RBC by directly measuring the number of electrical pulses corresponding to the red blood cells. The unit is 10^{12}/L.

b) **Hematocrit (Unit%)**

According to the RBC distribution histogram, hematocrit HCT is detected in unit of %.

Application scope of HCT: When the value is 0.0 – 33.3, the linear correlation coefficient is r ≥0.990, and the allowable deviation range is ± 1.0HCT when the value is 33.4 – 75.0 , the linear correlation coefficient is r ≥0.990, and the allowable deviation range is ± 3.0.

c) **Mean corpuscular volume**

$$MCV = \frac{(\overline{HCTX10})}{RBC}$$

d) Mean corpuscular hemoglobin concentration

$$MCH = \frac{HGB}{RBC}$$

e) Mean corpuscular hemoglobin concentration

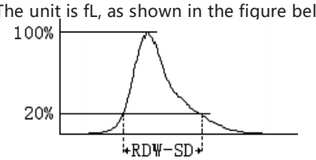
$$MCHC = \frac{HGB}{HCT} X100$$

f) **Red blood cell distribution width-coefficient of Variation**

The RDW-CV is obtained from the RBC histogram and is a coefficient of variation of the volume distribution expressed as a percentage.

g) **RDW-SD**

The RDW-SD is the width of the histogram at 20 relative to the peak of the RBC distribution histogram. The unit is fL, as shown in the figure below.



g) RDW-SD

The RDW-SD is the width of the histogram at 20 relative to the peak of the RBC distribution histogram. The unit is fL. as shown in the figure below.

PLT Sheath Flow Impedance Channel

a) **Number of Platelets**

The analyzer obtains the number of platelets PLT by directly measuring the number of electrical pulses corresponding to the platelets. The unit is **10^9 / L**

b) **Mean platelet Volume**

According to the platelet distribution histogram, the mean platelet volume MPV is calculated in fL.

c) **Platelet distribution width**

The platelet distribution width PDW is obtained from the platelet distribution histogram, which is the geometric standard deviation of the platelet volume distribution.

d) **Platelet hematocrit**

The analyzer calculates platelet hematocrit PCT by the following formula in %.

$$PCT = \frac{PLTXMP}{10000}$$

e)Platelet -Larger Cell Ratio (P-LCR)

P-LCR is the ratio of giant platelets obtained from a 12fL landmark or greater.

3.5.3 FCP Channel

The FCP channel is used to count platelets by fluorescent staining detection method, and the measurement parameter is PLT-F.

Application scope of FCP channel: When the value is ≥ 20 X 10^9/L, its precision is ≤ 5.0 When the value is ≥ 60 X 10^9/L, its precision is ≤ 3.0 When the value is ≥ 100 X 109/L, its precision is ≤ 2.5 .

When the value is 0 – 2000 X 10^9/L, the linear correlation coefficient is r ≥ 0.990, and the allowable deviation range is ± 5.0% or ± 10 X 10^9/L when the value is 2001 - 5000 X 10^9/L, the linear correlation coefficient is r ≥ 0.990,
and the allowable deviation range is ± 6.0 .

3.5.4 FCR Channel

a)Reticulocyte Ratio

$$RET = \frac{\text{Number of particles in the reticulocyte}}{\text{Number of particles in mature red blood cell region + number of particles in reticulocyte region}}$$

Application scope of reticulocyte ratio: when RET is between 1.00 – 4.00 and RBC ≥ 3.00 X 10^{12}/L, the precision of RET is ≤ 15.0 when RET is between 0.00-30.00 , the linear correlation coefficient is r ≥ 0.990, and the allowable deviation range is ± 0.30RET or ± 20.0 .

b)Number of reticulocytes

$$RET = \frac{RET\% \times RBC}{10000}$$

Application scope of reticulocyte number: When RET is 1.00 – 4.00% and RBC ≥ 3.00 X 10^{12}/L, the precision of RET is ≤ 15.0 when RET is 0 – 800 X 10^9/L, the linear correlation coefficient is r ≥ 0.990, and the allowable deviation range is ± 15.0 X 10^9/L or ± 20.0.

The accuracy of the number of reticulocytes RET is: 1 r ≥ 0.90; 2) ±20% or ±15X10^9/L.

c)Low Fluorescence Ratio

$$LFR = \frac{\text{Number of particles in LFR region}}{\text{Number of particles in the reticulocyte region}} x1000$$

d)Middle Fluorescence Ratio

$$MFR = \frac{\text{Number of particles in MFR region}}{\text{Number of particles in the reticulocyte region}} x1000$$

e) High Fluorescence Ratio

$$HFR = \frac{\text{Number of particles in HFR region}}{\text{Number of particles in the reticulocyte region}} x1000$$

f) Immature Reticulocyte Fraction

$$IRF = MFR + HFR$$

g)Reticulocyte hemoglobin concentration

$$RET\text{-}He = 5.5569 e^{0.01 \times RET \text{ Forward scattered intensity}}$$

3.5.5 HGB Measurement

The hemoglobin concentration HGB is calculated by the following formula and the unit is g/L.

$$HGB=Constant\times Ln\left(\frac{Background\ Transmission\ Light}{Sample\ Transmission\ Light}\right)$$



Chapter4 System Installation

4.1 Overview

The analyzer is strictly inspected before leaving the factory, and properly packed to ensure that it is protected from impact during transportation. When the analyzer arrives, please carefully check whether the packaging is intact. If there is any damage, please notify the after-sales service department of the Company or the agent in your region immediately.

Precautions

- Unpacking or implementing the installation process by personnel not authorized or trained by the Company may cause damage to the analyzer. Therefore, please do not unpack or install the analyzer without the authorized personnel of the Company.
- If the analyzer will not be used for a long time, please shut it down, empty the reagents inside the analyzer and rinse with deionized water （DI）. Then pack it properly. The storage conditions of the analyzer must meet the relevant storage requirements.

4.2 Installation Requirements

Before installing the analyzer, make sure to meet the following space, power, environmental, and electromagnetic compatibility EMC requirements.

4.2.1 Space Requirements

To meet the requirements for maintenance space of the analyzer, and to make sure sufficient heat dissipation and avoid squeezing the pipeline behind the analyzer, the installation position of the analyzer should meet the following requirements:

- The left&right doors of the analyzer should keep ≥100cm away from the walls.
- The back panel of the analyzer should keep ≥50cm away from the wall.
- Make sure there is enough space on the operation table and under the analyzer for diluent, lysing reagent, waste container, etc.
- Do not place the analyzer where the switch is difficult to turn off.

4.2.2 Power Supply Requirements

	Voltage	Frequency	Power	Fuse Specification
Main unit	AC 100V-240V	50Hz-60Hz	≤ 420VA	F5AL250V
Compressor unit	AC 110V/220V	60Hz/50Hz	≤ 450VA	F5AL250V

Our company could provide one of two compressor models that are called CP-10 or CP-11. The CP-10 is suggested to work under the condition of AC 220V- 50Hz/60Hz, however the CP-11 is suggested to work under the condition of AC 110V-117V- 60Hz. Please confirm the compressor model with the saleman before purchase.

⚠ Warning

- The analyzer must be used in a well-grounded condition.
- The analyzer must use fuses of the specified specifications.
- Confirm that the voltage meets the requirements.

4.2.3 Environmental Requirements

	Normal Working Environment	Storage Environment	Operating Environmen
Temperature	15°C -30°C	-10°C -40°C	5°C-40 °C
Relative Humidity	30% -85%	10%-90%	10%-90%
Relative Pressure	70kPa-106kPa	50kPa-106kPa	70kPa-106kPa
Altitude	≤ 3000m	≤ 3000m	≤ 3000m

- The environment where the analyzer is installed and used should be as clean as possible without dust, mechanical vibration, large noise or power interference.
- Do not place the analyzer near sources of strong electromagnetic interference, so as not to affect the normal operation of the analyzer.
- Keep away from brush-type motors, flashing fluorescent lights, and electrical contact devices that are frequently switched on or off.
- The analyzer should be protected from direct sunlight and kept away from heat sources.

⚠ Warning

- Do not use the analyzer in a flammable or explosive environment to avoid danger.
- Do not move the analyzer without permission. If you need to move it, please contact our service department.

4.2.4 Electromagnetic Compatibility (EMC) Requirements

To guarantee a proper EMC environment for the analyzer, pay attention to the following points:

- Keep the analyzer away from facilities with high electromagnetic radiation, such as dental beds, pacers, and welding equipment.
- Keep the analyzer away from high-power devices that are started up frequently, such as refrigerators and centrifuges.
- Keep the analyzer away from strong magnetic fields, such as radiology equipment.
- Do not connect too many devices to the power strip for the analyzer.

- During a test, do not plug in or out other devices that are connected to the same power strip as the analyzer.
- Make sure that the analyzer power socket is reliably grounded. Do not use non-standard power cable.

The Automatic Hematology Analyzer has passed the EMC test, and complies with the IEC 61326-1/E 61326-1 Electrical equipment for measurement, control and laboratory use – EMC requirements — Part 1: General requirements, IEC 61326-2-6/E 61326-2-6 Electrical equipment for measurement control and laboratory use - EMC requirements - Part 2-6: Particular requirements - In virtro diagnostic IVD medical equipment.

During use, please strictly follow these requirements. Otherwise, EMI may be caused to other devices, and the electromagnetic immunity or basic functions of the Automatic Hematology Analyzer may be deteriorated or even lost.

This device is designed and tested as a Class A device, Group I as specified in CISPR 11 . The fully automated analyzer can be used in all non-household facilities and all household facilities that are not directly connected to the public low-voltage power supply network of residential applications.

Portable and mobile RF communication devices may affect the normal operation of the Automatic Hematology Analyzer. Make sure that the distance from portable and mobile RF communication devices to the Automatic Hematology Analyzer meets certain requirements. For details, refer to Table 4.

Warning 1:Use only the transducer and cable sold by device or system manufacturer as spare parts for the internal components. The use of unauthorized accessories, transducer and cable may increase the device or system emission or decrease the immunity.

Warning 2: Do not keep the device or system near other devices. Do not stack the device or system with other devices. If this cannot be avoided, observe and check to make sure that the device or system is functional under such circumstances.

Warning 3: Do not use this device near a strong radiation source. Otherwise,

Table 1 declaration— electromagnetic emission

Emission test	Compliance
Conducted Emissions IEC/CISPR 11	Group 1
Radiated Emissions IEC/CISPR 11	Class A
Harmonic Current Emissions IEC 61000-3-2	Class A
Voltage Fluctuations and Flicker IEC 61000-3-3	Complies

Table 2 declaration— electromagnetic susceptibility

Immunity test	IEC 61326- -6 Test level guideline	Compliance level
Electrostatic Discharges ESD IEC 61000-4-2	±2 kV, ±4 kV contact discharge voltage ±2 kV, ±4 kV , ±8 kV air discharge voltage	±2 kV , ±4 kV contact discharge ±2 kV, ±4 kV, ±8 kV air discharge
Electrical Fast Transients EFT IEC 61000-4-4	±1 kV	±1 kV
Surge IEC 61000-4-5	±1 kV line to line ±2 kV line to earth	±1 kV line to line ±2 kV line to line
Voltage Dips, Interruptions, and Variations IEC 61000-4-11	0% UT* during 1 cycle 40%UT* during 5/6 cycles 70%UT* during 25/30 cycles 0%UT* during 250/300 cycles	0%UT* during 1 cycle 40%UT* during 5/6 cycles 70%UT* during 25/30 cycles 0%UT* during 250/300 cycles
Power-Frequency Magnetic Fields 50/60Hz IEC 61000-4-8	3 A/m	3 A/m

NOTE:UT is the a.c. mains voltage prior to application of the test level. 5 cycles, 25 cycles and 250 cycles for 50Hz 6 cycles, 30 cycles, 300 cycles for 60Hz.

Table 3 declaration electromagnetic susceptibility (continued table)

Immunity test	IEC 6 3 6-2 -6 test level	Compliance level
Conducted Disturbances IEC 61000-4-6	3V 150 kHz to 80 MHz	3V 150 kHz to 80 MHz.
Radio Frequency Electromagnetic Fields IEC 61000-4-3	3V/m 80 MHz-1 GHz 3V/m 1.4 GHz-2 GHz 1V/m 2.0 GHz-2.7 GHz	3V/m 80 MHz-1GHz 3V/m 1.4 GHz-2GHz 1V/m 2.0 GHz-2.7GHz

4.3 Connection of the Analyzer

Our authorized personnel complete the following connections when installing the analyzer.

4.3. Connection of Compressor unit

The connection between the analyzer and the compressor unit is shown in the following figure. Before use, the user must confirm whether the connection is in place and secure.

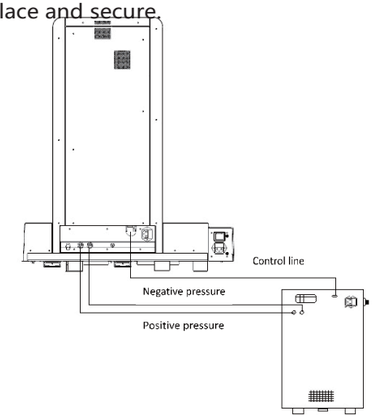


Figure4-1Connection of compressor

4.3.2 Connection of Pipeline

⚠ Warning

- Dispose of the reagent, waste liquid and waste sample accordance with regional and national regulations on the discharge and processing of materials with microbial infectivity such as reagent and waste.
- The reagent can irritate the eyes, skin and mucous membranes. Operators should observe laboratory safety regulations when using or contacting items related to reagents, and wear personal protective equipment such as laboratory protective clothing, gloves, etc.
- As soon as the reagent comes into contact with the skin, rinse immediately with plenty of water and seek medical attention if necessary.
- As soon as the reagent comes into contact with the eyes, rinse immediately with plenty of water and immediately get medical attention.

Precautions

- Do not place reagents or other liquids on top of the analyzer.

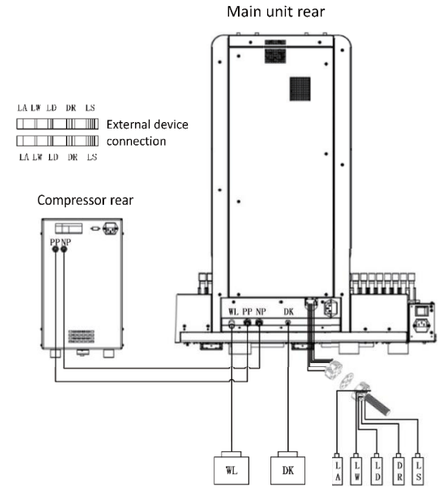


Figure 4-2 Pipeline connection of the analyzer

WL: Waste liquid port

DK: Diluent port

PP: Positive pressure port of compressor unit

NP: Negative pressure port of compressor unit

4.3.3 Connection of Optional Devices

The analyzer and optional devices should be installed in accordance with the manual.

Make sure that all lines are connected securely.

Precautions

- The printer, mouse, keyboard, and scanner are connected through the SB port of an external PC.
- The scanner recognizes the barcode of one-dimensional barcode type.
- The analysis module can be combined with the sample loader optional to form a stand-alone analyzer, or it can be combined with a specific track to form an analysis system. For details, please consult Agappe Customer Service Center.

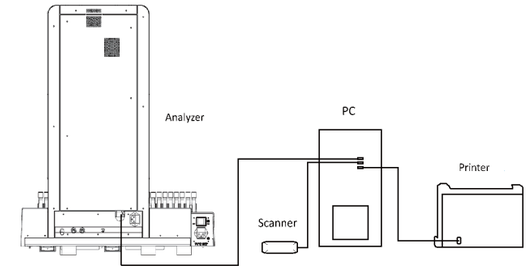


Figure 4-3 Connection of optional

4.4 Software Operating Environment

4.4.1 Hardware Configuration

- CPU : 4.0GHz or higher, quad-core or higher
- Memory: 4G or higher
- Hard Disk: 500GB (Enterprise Mechanical Hard Disk [Black Disk]) or higher
- Peripherals: Printer, mouse, keyboard, scanner
- Audio: The computer host or display must support audio playback
- NIC:

✓ Dual IC configuration, one of which must support Wake on LA

32-bit / 64-bit Windows7 operating system

64-bit windows10 operating system

4.4.2 Software Configuration

✓

- Operating System:

✓

- System software:

✓ The system should support Windows .NET Framework 4.4 and above

- Application

✓ Mysql Server 5.7

4.4.3 System environment

- User rights:

You must use administrator rights to operate the instrument software.

- Security software

Anti-virus software needs to be installed for necessary virus protection and system vulnerability patching. When necessary, engineers can use local storage media or network update methods to update security software and system vulnerabilities.

- Hibernation management

You need to set the computer hibernation, standby and other options to never hibernate.

- Firewall

Use Windows's own firewall and enable it by default. At the same time, set the security policy to allow the software to interact with the data through the firewall.

- User rights control

✓ Start Windows interactive login Disable

✓ Windows remote login

✓ Enable Windows administrator account status function

Precautions

- Copy the supporting installation package to the hard disk of the external PC computer then double-click the installation package on the hard disk, and follow the software installation prompts to complete the installation operation.

Chapter5 Basic Settings

5.1 Overview

The analyzer has been initialized before leaving the factory. The parameters and reference value ranges are set to the system defaults.
Users can reset some of these parameters according to different actual needs.

5.2 System settings

5.2.1 Management Settings

Click "Management Setting" to enter the management settings interface to manage DCS settings.

5.2.2 System

5.2.2.1 Shutdown settings

Click "Shutdown and sleep" to set whether to automatically shut down the software and computer after shutting down the analyzer.

5.2.2.2 Date format

Click "Date format" to set the date display format of the software as required.

5.2.2.3 User management

Click " ser Management" to enter the user management interface.

5.2.2.4 CSV output

Click "CSV output" to set the CSV output file format as required.

5.2.2.5 Security

Click "Security", you can set whether the output data and files contain patient information and patient information, and set the computer lock screen time.

5.2.3 Display

5.2.3.1 Scattergram

Click "Scatter diagram" to set the background color of the scattergram as required.

5.2.4 Connect

5.2.4.1 LIS settings

Click "LIS setting" to configure the connection with the LIS or the report management system host.

5.2.5 Auto Processing

5.2.5.1 Auto validate

Click "Auto validate" to set the method and sample conditions for auto validation.

5.2.5.2 Auto output

Click "Auto output" to set the method and sample conditions for auto output.

5.2.5.3 Analysis reservation

Click "Analysis reservation" to set the unique sample ID used for sample analysis as required.

5.2.6 Reference Range

5.2.6.1 Reference range

Click "Reference range" to set the reference value range of the analysis result.

5.2.6.2 Reference group settings

Click "Reference group settings" to set the classification of the reference range group.

5.2.6.7 Unit

Click " Unit" to enter the unit setting interface. Set the parameters to use different units.

5.2.6.8 QC

You can set controls, control alarms, control table fixed comment, and control data auto output.

5.3 Instrument settings

5.3.1 Management Settings

Click "Management setting";you can enter the management settings interface

5.3.2 Alarm Sound

Click "Alarm Sound" you can set the alarm type, alarm error, and measurement stop error.

Chapter 6 Daily Operations

6.1 Overview

This chapter introduces the daily operations of the analyzer from power-on, sample analysis to power-off.

The analyzer supports two analysis modes: manual loading and autoloading.

It supports four sample modes: WB, LW, PD, and BF. It supports 12 detection modes, including CBC, CBC + DIFF, CBC + RET *, CBC + DIFF + RET *, CBC + PCF *, CBC + DIFF + PCF *, CBC + RET * + PCF *, CBC + DIFF + RET * + PCF *, CBC + DIFF + AWS *, CBC + DIFF + AWS * + RET *, CBC + DIFF + AWS * + PCF *, and CBC + DIFF + AWS * + RET * + PCF *.

*This content is not displayed on all analyzer models.

Analysis Mode and Sample

nalysis Mode		Sample Mode
Autoloading*1		WB mode, LW mode
Manual loading	General specifications	WB mode, LW mode, BF mode
	Micro blood collection * 2	WB mode, LW mode, PD mode, BF mode
	Micro blood collection * 3	WB mode, LW mode, PD mode, BF mode

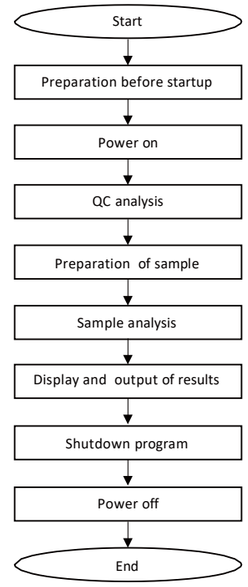
* 1: Auto loading: Measurement mode for auto transfer of tube rack
* 2: In order to reduce the minimum volume also called the dead volume , open the blood collection tube cap for manual measurement.
* 3: Micro measurement micro blood collection tube measurement : Measurement using a micro blood collection tube and measurement with the cap open.

Sample Mode and Detection

Detection mode		Sample Mode			
		WB	LW	PD	BF
1	CBC	√	–	√	–
2	CBC+DIFF	√	√	√	–
3	CBC+RET*	√	–	–	–
4	CBC+DIFF+RET*	√	√	√	–
5	CBC+PCF*	√	–	–	–
6	CBC+DIFF+PCF*	√	√	√	–
7	CBC+RET*+PCF*	√	–	–	–
8	CBC+DIFF+RET*+PCF*	√	√	√	–
9	CBC+DIFF+AWS*	√	√	–	–
10	CBC+DIFF+AWS*+RET*	√	√	–	–
11	CBC+DIFF+AWS*+PCF*	√	√	–	–
12	CBC+DIFF+AWS*+RET*+PCF*	√	√	–	–

*This content is not displayed on all analyzer models

Daily Operation Process



Preparation before startup

Before turning on the power of the analyzer, please check whether the waste container, reagents and connection lines, printer, analyzer and network communication connection are normal, etc. to ensure that the system can operate normally.

- Samples, calibrators, controls, waste liquid, etc., and areas in contact with these substances are potentially biologically infectious. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment such as protective clothing and gloves.
- There are fluidics inside the analyzer. You are recommended to check for liquid leakage during each self-test operation. If so, please immediately shut down the analyzer and contact our after-sales department.
- In order to ensure that the analyzer is in a safe state before use, please confirm whether the power supply, grounding conditions, waste discharge pipes, etc. meet the requirements of this manual.

⚠ Warning

- Dispose of the reagent, waste liquid and waste sample in accordance with regional and national regulations on the discharge and processing of materials with microbial infectivity such as reagent and waste.
- As soon as the reagent comes into contact with the eyes of the operator, rinse immediately with plenty of water and immediately get medical attention.
- The operator keeps a certain distance from the moving parts of the analyzer to prevent being pinched by the moving parts.

Precautions

- The operator should use the supporting reagents of the Company, and use and store them strictly in accordance with the reagent instructions.

6.2 Startup

Start the computer and open the analyzer control software:

1.Turn on the analyzer supporting computer and display

2.After accessing the operating system, double-click the analyzer control software icon to run the software

3.After the software is started, enter the login interface

4.The operator enters the correct user name and password, and clicks"Login" to enter the software main interface. If it has been set to "Auto login", it is not necessary to enter your username and password again.

Start the analyzer:

1. Set the power switch on the back of the compressor unit to I
2. Set the power switch on the back of the sample loader to ▮ the power indicator is on
3. Press the power button at the top left of the analyzer to start the analyzer the power indicator is on, and the status indicator on the analyzer is on
4. The analyzer performs a power-on self-test. The power-on self-test includes initialization of mechanical parts, cleaning, blank count detection, and temperature detection.
5. After the power-on self-test is completed, the analyzer enters the standby state.

Precautions

- The time required for the power-on self-test varies according to the previous shutdown situation.
- If the first blank count test result exceeds the blank range, the analyzer will automatically perform another blank count test, up to 3 times. If the three blank count detection results are not within the blank count range, it will be regarded as a blank count detection error.
- If a blank count detection error occurs during the power-on self-test, the analyzer will give a fault alarm prompt, and will perform the "Auto Flush" processing sequence.
- If you cannot eliminate the error by following the above procedure, please contact our service department.
- See the appendix for the blank count range of each parameter.
- If you perform sample analysis when "Blank check error" exists, you cannot obtain reliable analysis results.
- During blank detection, the blank detection sample No. is [BACKGROUNDCHECK]

6.3 Quality Control

Prior to sample analysis, daily quality control of the analyzer is required to ensure reliable analysis results. For specific quality control analysis and operation methods, refer to Chapter 7 Quality Control.

6.4 Analysis Mode

The analyzer supports the following two analysis modes:

Autoloading mode: In this operation mode, the operator places the sample to be tested in the sample rack also called tube rack , and the analyzer automatically mixes the sample tubes in sequence and pierces and aspirates the sample for testing.

Manual loading mode: In this operation mode, the operator manually mixes the samples and places them in the manual loading position. The analyzer automatically pierces and aspirates the sample for testing.

6.5 Preparing Samples

- Samples, calibrators, controls, waste liquid, etc., and areas in contact with these substances are potentially biologically infectious. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment (such as protective clothing and gloves) .

PC Warning

- Do not touch the blood samples of patients

6.5.1 WB Sample

Mix the venous blood with the anticoagulant K2EDTA, and draw a certain amount of blood according to the instructions of the tube used. After collecting the samples, be sure to perform the analysis within 4 hours. If the sample cannot be analyzed within 4 hours, store it at 2 C to 8 C until analysis is possible. Before analyzing refrigerated samples, wait the sample to return to

room temperature and mix it thoroughly before sample analysis.

6.5.2 PD Sample

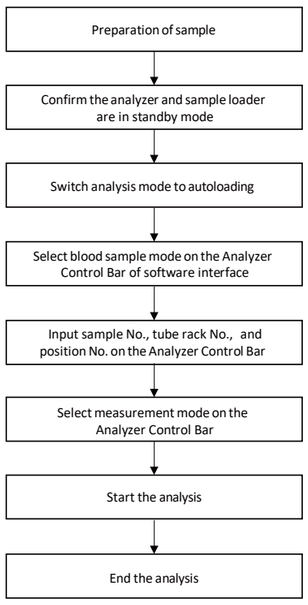
Pre-dilute peripheral and venous blood for 7-fold dilution before sample analysis. Pre-diluted samples can only be diluted outside the analyzer.

6.6 Registering Worksheet Sample

Before the measurement, it is necessary to register a worksheet sample, and enter the relevant detection information according to the content displayed on the interface. For specific worksheet registration, see the subsequent sections of this chapter.

6.7 Autoloading nalysis

Autoloading analysis start supports two methods: manual start and auto start. The operation flow is as follows:

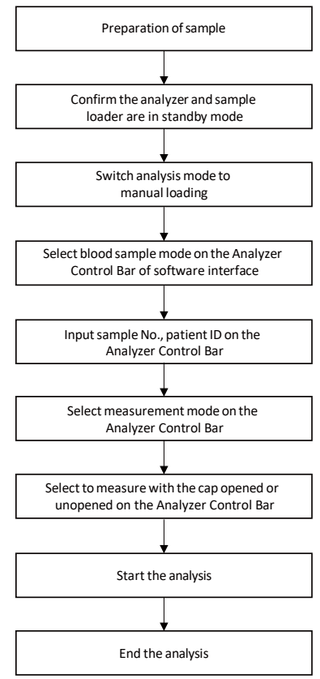


Aborting Autoloading Measurement:

Click the autoloading measurement button in the analyzer control bar a dialog box will pop up select [Yes] in the dialog box.

6.8 Manual Loading Analysis

The operation flow is as follows:



The user can set whether to enable barcode ID reader, analyzer blood aspiration sensor, and main unit query function through the analyzer settings. If the operator chooses to read the barcode ID when entering the sample information, the sample number can only be the scanned barcode ID.

6.9 Inserting Emergency

In the process of autoloading, if you need to insert emergency samples manual loading method, please do as follows:

1.Click the Mode Switch button on the analyzer to switch from autoloading mode to manual loading mode, and pause the autoloading measurement after the current sample measurement is completed

2.After the autoloading measurement is suspended, follow the manual loading procedure to insert emergency sample information

3.Shake the sample to be tested and place it in the manual loading chamber

4.Click the Start Measurement button on the analyzer to start the emergency measurement.

6.1.0 Recovering Autoloading Measurement

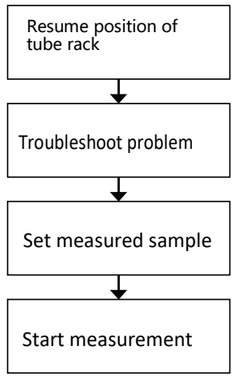
When the measurement is interrupted due to the insertion of an emergency sample during the autoloading process, follow the steps below to resume autoloading.

1.Click the Mode Switch button on the analyzer to switch from manual loading mode to autoloading mode after the emergency sample measurement is completed

2.The analyzer starts measurement and resumes measurement from where it was paused.

6.11 Handling Downtime Failure During Autoloading

During the autoloading process, the analyzer will stop the measurement if downtime failure occurs.



6.12 Worksheet

6.12.1 Overview

Click "Worksheet" to open the worksheet interface. Then register, modify and view the sample.

6.12.2 Toolbar

Includes the Register, Modify, Download, Filter, Sort, Output, p, Down, Find, To Be Tested, File, Delete and other function buttons.

- Register

  The operator can input a work order through "Register".

- Modify

The operator can modify the stored worksheet to be tested through "Modify".

- Download

  The operator can download worksheets from the main unit.

- Filter

  The operator can set different filter conditions so that work orders are displayed according to the set filter conditions.

- Sort

  The operator can set different arrangement conditions so that work orders are displayed according to the set arrangement conditions.

- Output

  The operator can print the selected work order on paper.

- Up/Down

  By clicking " p", the selected work order in the worksheet list will move to the previous one. By clicking "Down", the selected work order in the worksheet list will move to the next one.

- Find

  By clicking "Find", users can find the specified work order by setting conditions.

- To be tested

  Only the work orders to be tested are displayed in the worksheet list.

- File

  The operator can back up and restore all work orders in the worksheet.

- Delete

  The operator can delete the selected work orders.

Precautions

- The sample No. of the sample to be measured cannot be repeated, and the occupied sample No. will be released only after the measurement is completed.
- The patient ID is unique. The same patient ID cannot have different patient information at the same time. The system uses the patient ID to identify the patient.
- Only the worksheet samples to be measured can be modified.
- When the sample analysis keyword is set to the sample No., "Download" cannot be used.
- If the filter condition is set to "To be tested", other filter conditions cannot be used, that is, "Filter" cannot be used.

6.12.3 Worksheet List

The worksheet list area displays the stored work orders. The displayed contents include the analysis parameter list, patient information list, and order information list. Up to 2000 work orders can be stored.

6.12.4 Font Size Settings

The operator can set the size and line height of the displayed content in the

order information list as needed, and click different font sizes to change the size and line height displayed.

6.13 Result details

6.13.1 Overview

The system automatically saves the analysis results of the detection to the database every time.

The system database can store the information of 200,000 samples, which may be different according to the analyzer configuration. Each sample information contains 2 histograms RBC, PLT , 5 scattergrams FCW, FCD, FCR *, FCP *, FCA * , 37 * blood reportable parameters and 34 * blood research parameters and results hints. BF detection output 1 histogram RBC , 2 scattergrams FCD, FCR , and 17 BF parameters.

Users can choose to view history data in batches, or view detailed analysis results in detailed browsing mode display histograms and scattergrams .

*This content is not displayed on all analyzer models.

6.13.2 History Data Mode

Click the "History" icon to enter the history data interface.

6.13.2.1 Viewing Results

Users can view the results saved by the system on this interface, and scroll the pages up and down to view the results of different pages.

6.13.2.2 Selecting Results

- Selecting Single Result

The user can arbitrarily select the analysis results to be viewed or further operated according to the needs left-click or press the ↑or ↓arrow on the keyboard , and the selected results are highlighted.

- Selecting Multiple Results

The user can continuously select the analysis result by holding down the left mouse button and dragging in the result list. When you need to select the results by segments, hold down the ctrl key on the keyboard while selecting the results as required.

Precautions

- The results selected in batches are highlighted, and the result display column only displays the samples indicated by the leftmost list arrow.

6.13.2.3 Validating Results

You can validate the sample results in the history data interface.

Precautions

- The sample No. that starts with "QC-" indicates the quality control data, the sample NO. that starts with "BACKGROUNDCHECK" indicates the blank count detection data, and the sample NO. that starts with "CAL- MEA" indicates the calibration data. Users are not recommended to modify the quality control data, blank count data, and calibration data.
- The sample No. that starts with "QC-" indicates the quality control data, the Sample No. that starts with "BACKGROUNDCHECK" indicates the blank count detection data, and the sample No: that starts with "CAL-MEA" indicates the calibration data, all of which can be deleted.
- The validated status is indicated by a "V" in the validated column.
- Only the validated samples can be printed. You can cancel validated for the validated samples, and you can also cancel validated for the printed samples.
- When the filter condition of "Last 20" is enabled, the sample cannot be reviewed.

6.13.2.4 Modifying samples

The user can modify the sample in the history data interface.

Precautions

- The samples in the validated state cannot be modified.
- When the filter condition of "Last 20" is enabled, the sample cannot be modified.

6.13.2.5 Deleting samples

Users can delete data that they do not want to keep as needed.

Precautions

- When the filter condition of "Last 20" is enabled, the sample cannot be deleted.

6.13.2.6 Uploading results

Select the results you want to upload to the LIS, and click "Output-Computer", and the data will be automatically uploaded to the LIS server.

Precautions

- To use the output function, you should successfully connect the analyzer to LIS.
- You can enable "Automatic output" in "System settings", and the new sample will be uploaded to the server automatically after the test is completed.
- When the filter condition of "Last 20" is enabled, the sample cannot be uploaded to the LIS server.

6.13.2.7 Printing reports

Users can print out the validated results in the history data interface.

Precautions

- Before performing a report printout, confirm the printer installation, connection status, and adequacy of paper supplies.
- Unvalidated samples do not allow print preview or printout.
- When the filter condition of "Last 20" is enabled, the report cannot be

6.13.2.8 Backing up results

The user can back up the selected result data.

6.13.2.9 Restoring results

Users can restore the data that has been backed up.

Precautions

- When the filter condition of "Last 20" is enabled, the result cannot be restored.

6.13.2.10 Exporting results

Users can export the results to the computer in CSV format individually or in batches in the history data interface for sample analysis using statistical tools such as EXCEL.

Precautions

- The file name is automatically generated by the system, and the user can also rename it as needed.
- The save format is CSV.
- When the filter condition of "Last 20" is enabled, the result cannot be

6.13.2.11 Filtering results

Users can set conditions to filter the history sample results that meet the conditions.

Precautions

- With the filtering function, you can select single condition query or multiple condition combination query.
- When the filter condition of "Last 20" is enabled, other filter conditions

6.13.3 Detailed Browsing Mode

Click the "Result details" button the Result Details interface is displayed.

6. 13.3.1 Viewing Results

The result detail interface displays the analysis result of the currently selected sample, a negative/positive alarm, a retest action, a validated status, sample information, patient information, fault rule information, etc.
Users can click the up/down buttons in the toolbar to view the results, or use the ↑ /↓ key in the keyboard to switch.
Click "Custom param." to display the laboratory-specific interface. The laboratory-specific interface is used for experimental research. This interface can display research parameters in addition to reportable parameters.

Precautions

- Double-click the graph of the sample 2 histograms and 5 scattergrams to zoom in. After zooming in, you can also drag the mouse to zoom in again.

6.3.3. Operating instructions for result detail interface

Users can modify, validate, delete, and view unvalidated samples in the results detail interface. Users can view, print, export to LIS, and delete the validated samples in the result detail interface.

Precautions

- The sample No. that starts with "QC-" indicates the quality control data, the sample No. that starts with "BACKGRO DCHECK" indicates the blank count detection data, and the sample No. that starts with "CAL-MEACAL" indicates the calibration data. Users are not recommended to modify the
- quality control data, blank count data, and calibration data.
- The sample No. that starts with "QC-" indicates the quality control data, the sample No. that starts with "BACKGRO DCHECK" indicates the blank count detection data, and the sample No. that starts with "CAL-MEACAL" indicates the calibration data, all of which can be deleted.
- The validated samples are marked with "Validated" in the upper left corner and displayed in green, and the unvalidated samples are marked with a gray background.
- Only the validated samples can be printed. You can cancel validated for the validated samples, and you can also cancel validated for the printed samples.
- When the filter condition of "Last 20" is enabled, the sample cannot be
- reviewed.

6.3.3.3 Last 20

Click the "Last 20" button to control whether the latest 20 samples are displayed in the result details interface when the latest samples are stored in the database.

6.14 Shutdown and Exit Program

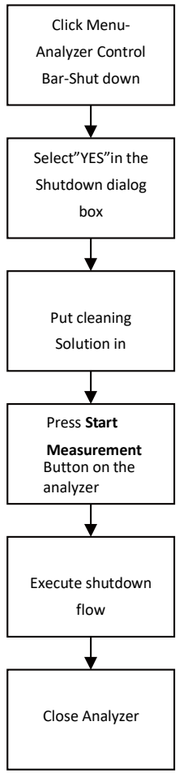
If the analyzer is on, you must perform a shutdown operation at the end of each day of analysis or after 24 hours of continuous operation.
When the analyzer is being shut down, the analyzer tubing and analysis components will be cleaned.
If the analyzer should be unused for a long period of time, please contact our service department.

- Samples, calibrators, controls, waste liquid, etc., and areas in contact with these substances are potentially biologically infectious. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment (such as protective clothing and gloves) .

Precautions

- To ensure reliable and stable results of the analyzer, be sure to perform the shutdown procedure as below.

The shutdown procedure is as follows:



Chapter7 Quality Control

7.1 Overview

In order to monitor the performance of the analyzer during use, regular quality control is necessary. The laboratory should develop a set of standardized quality control operation procedures and strictly implement them.

After the analyzer has been calibrated or major components have been replaced, additional quality control operations are required.

- Calibrators and controls are just as potentially biologically infectious as samples. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment such as protective clothing and gloves.

Precautions

- Only use the specified control products, and store and use them strictly in accordance with the requirements of the control manual.

7.2 QC Material

When performing X-bar * or L-J quality control, control measurement is required. The QC materials are divided into three levels: Level 1, Level 2 and Level 3.

*This content is not displayed on all analyzer models

Precautions

- Use only the specified controls. Controls can help to determine the status of the analyzer.

7.3 QC Type

The analyzer QC types include L-J control, X-bar * control and X-barM control.

L-J control: Use the data of a single control test as control data.

X-bar* control: Measure the control twice consecutively every time, and use the average value as control data.

X-barM control: Calculate the weighted average of the measured data of patient negative samples as control data. The number of negative samples are usually 20 (the number of samples can be set).

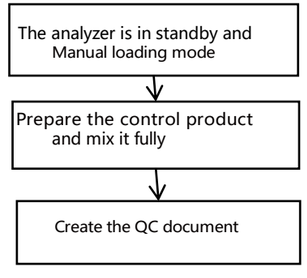
Precautions

- The user can set the control type to L-J or X-bar * through "Control Setting" in software settings and set the number of samples in X-barM for calculating the weighted average.
- Users can set whether to enable X-barM through the Analyzer Control Bar.

*This content is not displayed on all analyzer models

7.4 QC Preparation

The QC preparation process is as follows:



7.5 QC Document

7.5.1 Registering QC Document

Before performing control tests, you must register the QC documents. Input the QC document as follows:

1. Click the "QC" icon to access the QC interface.
2. Select a blank QC document, and the interface displays the corresponding document information.
3. Click "Register", and enter the batch o., expiration date, labeling value (ie target value) and limit range provided in the manual.
4. After completing the input, click the "OK" button to save this QC document information.

Precautions

- The registered document No. cannot be registered again and can only be modified.

7.5.2 Modifying QC Document

Click the "QC" icon to enter the QC interface, select the QC document No. to be modified, and click "Modify" to modify the relevant information of the QC document.

7.5.3 Deleting QC Document

Select the QC document to be deleted and click the "Delete" icon to delete the corresponding document.

Precautions

- The "Delete" icon is not enabled when the selected document is a blank QC document.

7.6 Implementing QC Analysis

The QC method can be performed by the loading method supported by the instrument.



- Controls are potentially biologically infectious. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment (such as protective clothing and gloves) .

Follow the steps below for QC analysis.

1. Check if there is a fault prompt at present. If so, eliminate the fault first

   a If the analyzer status indicator is off or of other colors, there may be a fault b If the analyzer status indicator is green, you can perform a control test.
2. Confirm that the analyzer is in manual loading mode and in standby state
3. Put the prepared control in the tube holder. Mix the control thoroughly before putting it in
4. Enter the analysis control bar, click "Menu", select "QC analysis" to enter the QC analysis interface
5. Select the control information for QC analysis and click "OK" to enter the measurement interface
6. Press the Start Measurement button on the analyzer to start the control measurement
7. After the analysis is over, the obtained control results will be displayed on the current interface. Click "OK" to automatically save them to QC.

Precautions

- When performing X-bar QC analysis, repeat step 6 above for 2 measurements.
- During the QC analysis, you can click "Cancel" to abort the QC analysis.

7.7 QC Chart

You can enter the QC chart interface in the following ways:

In the QC interface, select the document No. where the chart is to be viewed, and click "QC List".

On the QC list, the selected QC document data will be displayed as a time series line chart.

7.7.1 Viewing QC Analysis Results

After the QC analysis is completed, the analysis results will be automatically displayed on the QC acceptance interface.

Precautions

- Regardless of whether the QC analysis results are accepted or not, they will be displayed in the History and Result details interfaces.

Chapter 8 Calibration of the Analyzer

8.1 Overview

The purpose of calibrating the analyzer is to ensure the accuracy of the analysis results.

The analyzer provides auto calibration and manual calibration.

You can calibrate all parameters in WBC, RBC, HGB, HCT, and PLT.

8.2 Calibration Frequency

The analyzer is calibrated before leaving the factory.

The accuracy of the analysis results can be guaranteed for a certain period of time after the analyzer is calibrated, but the analyzer should be calibrated in the following cases.

- Before using the analyzer again after the analyzer is not used for a long time
- After replacing the main parts
- Observe the control data. When a systematic deviation exists in the control data and cannot be corrected after repeated confirmation.

8.3 Calibration Mode

8.3.1 Calibration Preparation

Please confirm the following cases before calibration:

- The blank count of the analyzer meets the requirements. For the blank count requirements of the analyzer, see Appendix A. If the blank count is abnormal, please refer to the troubleshooting procedure to resolve it. If it cannot be resolved, please contact our service department
- The reagent balance is sufficient for this calibration. If there is insufficient reagent in the calibration process, the analyzer should be recalibrated
- The precision of the analyzer meets the requirements. For the precision requirements of the analyzer, see Appendix A. If the repeatability is abnormal, please refer to the troubleshooting procedure to resolve it. If it cannot be resolved, please contact our service department.

- Calibrators and controls are just as potentially biologically infectious as samples. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment (such as protective clothing and gloves).

Warning

- Because the sample probe is thin, and contaminated by calibrators, controls, and samples potentially biologically infectious, please pay attention during operation.
- Dispose of the packaging of discarded calibrators and controls in accordance with regional and national regulations on the discharge and processing of materials with microbial infectivity such as reagent and waste.

Precautions

- Only the user with administrator rights can perform calibration operations.
- The user should use the control and calibrator specified by the Company, and store them strictly in the storage environment specified by the instructions for calibrator, control and reagent.

8.3.2 Auto Calibration

Follow the steps below for auto calibration:

1. Check if there is a fault prompt at present. If so, eliminate the fault first
   a) If the analyzer status indicator is off or of other colors, there may be a fault
   b) If the analyzer status indicator is green, you can perform a calibration test.
2. Prepare calibrators or fresh blood samples that have been measured on a reference analyzer to obtain reference values
3. Confirm that the analyzer is in manual loading mode and in standby state
4. Put the prepared calibrator into the tube holder. Even if the analyzer will automatically mix the calibrator, the user still needs to mix it manually before calibration
5. Enter the analysis control bar, click "Menu", select "Calibration" , select "Corrector correction" to enter the calibrator calibration interface
6. Press the Start Measurement button on the analyzer to start the calibration measurements for 11 times consecutively
7. After the analysis is over, the obtained calibration result will be displayed on the current interface. When the calibration result meets the precision requirements of the analyzer, click "Calibration" to enter the calibration data confirmation interface
8. After entering the target value of the calibrator in the calibration data confirmation interface, observe that the Range Value, Error Rate, and Latest Compensation Rate of the calibration results meet the requirements. When all of them meet the requirements, click "OK" to enter the calibration coefficient confirmation interface
9. The user can manually change the latest compensation rate value on this interface, select the parameters to be calibrated, click "OK" to save the latest compensation rate, and update the compensation rate of the parameter selected for calibration
10. Apply the following formula to calculate the new calibration coefficient.

$$\text{Latest compensation rate} = \frac{\text{(current compensation rate * reference value)}}{\text{(measured average)}}$$

Precautions

- If the calibration result does not meet the specifications, there will be corresponding prompt information. Follow the inspection instructions.

8.4 Viewing Calibration Results

After the calibration analysis is over, the results saved in the analysis will be displayed on the Calibration interface.
Click "Calibration" on the main interface to enter the calibration interface.
On the calibration interface, all saved calibration records and single calibration measurement result corresponding to the calibration record will be displayed. The calibration interface can display up to 10 records. Records after the 10th will overwrite the existing records, starting from the earliest record. Calibration records can be viewed, exported, backed up, restored and deleted.

Precautions

- Regardless of whether the calibration measurement is over, the results will be displayed in the History and Result details interfaces.

Chapter 9
User Service

9.1 Overview

To ensure that the analyzer operates accurately and effectively, the operator should perform routine maintenance on the analyzer in accordance with the requirements of this chapter. The analyzer provides a variety of maintenance functions to facilitate the maintenance operations by the operator.

- The surfaces of all parts of the analyzer are potentially biologically infectious, and safety precautions should be taken during operation and maintenance.
- During the operation, if blood, or control, etc. spills on the analyzer surface, it can be wiped with disinfectant alcohol. During the cleaning process, safety protection measures should be taken, and the laboratory's safe operating practices must be strictly followed.

Precautions

- Improper maintenance may cause damage to the analyzer. The operator must perform maintenance in accordance with the manual.
- If you encounter a problem that is not clear in the manual, please contact our service department, and the Company's designated professional will give maintenance suggestions.
- The analyzer must be maintained using parts provided by the Company. If you have any questions, please contact our service department.

9.2 Maintenance

9.2.1 Liquid Path Initialization

Click the "Liquid Path Initialization" icon, the analyzer performs the initialization sequence for the fluidics and mechanical parts.

9.2.2 System Cleaning

Click the "Machine Cleaning" button, the analyzer will perform the cleaning sequence.

9.2.3 Aperture Unclogging

Click the "Jewel Hole Declogging" icon, the analyzer will perform the unclogging sequence.

9.2.4 Bubble Removal in Flow Chamber

Click the "Flow Chamber Bubble Remove" icon and the analyzer performs the bubble removal sequence in flow chamber.

9.2.5 Pressure Self-Test

Click the "Pressure Self-Test" icon, the analyzer will perform pressure

detection and return the real-time pressure value.

9.2.6 Reagent Guidance

Click the "Reagent Guidance" icon to enter the reagent introduction interface. The reagent will be introduced after the reagent is replaced or when there are air bubbles in the liquid path component.

9.2.7 Reservoir Self-Test

Click the "Reservoir Self-Test" icon, the analyzer will execute the Reservoir self- test sequence.

9.2.8 Waste Container Self-Test

Click the "Waste Chamber Self-test" icon, and the analyzer will perform the waste container self-test sequence.

9.2.9 Aspiration Syringe Self-Test

Click the "Sampling Syringe Self-test" icon, the analyzer will execute the aspiration syringe self-test sequence.

9.2.10 FCM Syringe Self-Test

Click the "FCM Syringe Self-test" icon, and the analyzer will perform the FCM syringe self-test sequence.

9.2.11 RBC Syringe Self-Test

Click the "RBC Syringe Self-test" icon, the analyzer will execute the RBC syringe self-test sequence.

9.2.12 Sample loader Self-Test

Click the "Autosampler Self-test" icon, the analyzer will execute the sample loader self-test sequence.

9.2.13 Horizontal Motor Self-Test

Click the "Horizontal Motor Self-test" icon, the analyzer will execute the horizontal motor self-test sequence.

9.2.14 Vertical Motor Self-Test

Click the "Vertical Motor Self-test" icon, the analyzer will execute the vertical motor self-test sequence.

9.2.15 Vortexer Up-Down Motor Self-Test

Click the "Catcher p-down Motor Check" icon, the analyzer will execute the vortexer up/down motor self-test sequence.

9.2.16 Vortexer Front-Rear Motor Self-Test

Click the "Catcher Front-rear Motor Check" icon, the analyzer will execute the vortexer front/rear motor self-test sequence.

9.2.17 Temperature Self-Test

Click the "Temperature Self-test" icon, the analyzer will perform temperature detection and return the real-time temperature value.

9.2.18 Launch of STAT Position

Click the "Launch of STAT Position" icon, the analyzer will perform the process of pushing the tube seat out of the STAT room.

9.2.19 STAT Position Advancement

Click the "STAT Position Advancement" icon, the analyzer will perform the process of pushing the tube seat into the STAT room.

9.2.20 Counter

Click the "Counter" icon, the analyzer will display the number of punctures performed by the puncture needle and the working hours of the laser.

9.3 Reagent Replacement

In the analyzer control bar, you can enter the reagent replacement interface, where you can perform the following reagent replacement:

- Automatic Hematology Analyzer DK Diluent
- Automatic Hematology Analyzer DR Diluent
- Automatic Hematology Analyzer LS Lysing Reagent
- Automatic Hematology Analyzer LW Lysing Reagent
- Automatic Hematology Analyzer LD Lysing Reagent
- Automatic Hematology Analyzer LA Lysing Reagent
- Automatic Hematology Analyzer FW Fluorescent Dye
- Automatic Hematology Analyzer FD Fluorescent Dye
- Automatic Hematology Analyzer FR Fluorescent Dye
- Automatic Hematology Analyzer FP Fluorescent Dye
- Automatic Hematology Analyzer FA Fluorescent Dye

Replace the diluent and lysing reagent as follows:

1. Enter the reagent replacement interface, click the reagent to be replaced, enter the barcode input interface, enter the corresponding reagent code, and confirm and save the barcode
2. Return to the previous interface, click the "OK" icon, the reagent replacement will start, and the reagent replacement progress will be displayed
3. After the replacement is completed, the current reagent balance will be displayed
4. If necessary, you can continue to replace other reagents as described above.

Replace the dye solution as follows:

1. Enter the reagent replacement interface and prepare new unexpired dye solution

2. Open the upper cover of the analyzer and move it to the highest point to prevent it from moving downwards

Precautions

- Do not open the upper cover of the analyzer during analysis and maintenance.

3. Pull up the lid of the dye solution vial to be replaced, and pull it to the top
4. Remove the old dye solution vial and put the new dye solution vial in
5. Pull the lid of the dye solution vial back to its original position
6. Close the upper cover of the analyzer, and the dye solution replacement will start automatically.



Chapter 10 Troubleshooting

10.1 Overview

This chapter introduces the possible failures of the analyzer during use and provides corresponding solutions.

- Samples, calibrators, controls, and wastes are potentially biologically infectious. During operation, be sure to comply with safe operation requirements of the lab and wear proper personal protective equipment (such as protective clothing and gloves).

Precautions

- This manual is not a maintenance manual, and only provides the measures that the operator should take when the analyzer fails.

10.2 Fault Information and Troubleshooting

During the use of the analyzer, if an abnormal condition is detected, the software will automatically pop up a fault dialog box, accompanied by an alarm sound.
The fault dialog will provide the name of the fault and its help information. The fault names are displayed in the order in which they occur.
The operator can click the fault name in the fault interface to make it selected highlighted , and view the processing information of the selected fault in the "Error dealing" box. The first fault information is selected by default.
When multiple faults occur at the same time, eliminate the highest priority fault first, and then eliminate the low priority fault to ensure that the fault can be eliminated correctly.
When a fault occurs, do not open the analyzer at will.

The following functions will be provided in the fault interface:

- Alarm Sound Elimination

  Click the "Stop beeping" icon to stop the fault alarm sound.

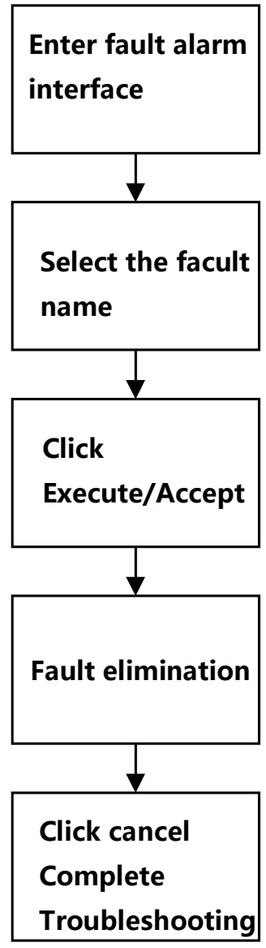
- Eliminating Faults

  Click the "Execute/Accept" button and the software will automatically eliminate the currently selected faults. For faults that cannot be eliminated automatically, the operator can perform corresponding troubleshooting information according to the information displayed in Troubleshooting.

- Closing the fault interface

  Click the "Cancel" button the fault interface will be closed, but the fault button still exists. Click the fault button again you can open the fault dialog box again. The fault

Eliminate the fault as follows:



If you cannot eliminate the error by following the above procedure, please contact our service department. If the laser fails, no repair is required, please contact our service department for replacement. If you do not use, control or adjust the device or perform the operation in each step in accordance with this

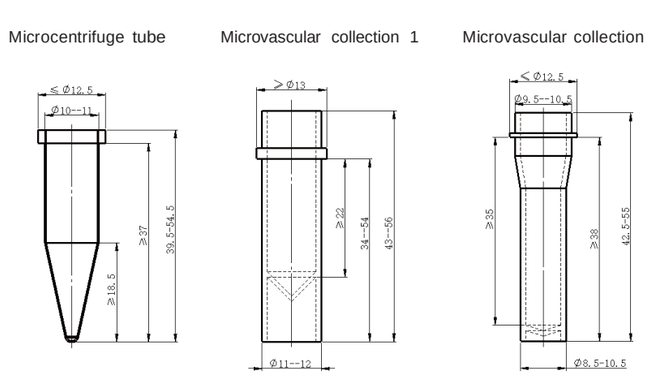
Appendix

1 . Vacuum Blood Collection Tube Model

The tubes in the following size supports autoloading: diameter 11mm-15mm (excluding tube cap) x length 70mm-85mm with tube cap .

The tubes in the above size range are verified to be applicable, but the possibilities still exist that there are adaptation problems with a few heterosexual tubes. If the tubes cannot adapt, please contact Agappe Customer Service Center.

2. Size of Micro Blood Collection Tube



- The above sizes do not include the cap and the corresponding extension tube.
- When the instrument is testing samples, open or remove the cap.

4. Parameter Description

Reportable Parameters (37)

Parameter	Parameter (English Name)	Default Unit
WBC	White blood cell count	10^9/L
BASO#	Number of basophils	10^9/L
BASO%	BASO	%
NEUTR#	Number of neutrophils	10^9/L
NEUT%	NEUT%	%
EO#	Number of eosinophils	10^9/L
EO%	EO%	%
LYMPH#	Number of lymphocytes	10^9/L
LYMPH%	LYMPH%	%
MONO#	Number of monocytes	10^9/L
MONO%	MONO%	%
IG#	Number of immature granulocytes	10^9/L
IG%	IG%	%
RBC	Number of red blood cells	10^{12}/L
HGB	Hemoglobin concentration	g/L
MCV	Mean corpuscular volume	fL
MCH	Mean corpuscular hemoglobin concentration	pg
MCHC	Mean corpuscular hemoglobin concentration	g/L
RDW-CV	Red blood cell distribution width-coefficient of variation	%
RDW-SD	RDW-SD	fL
HCT	Hematocrit	%
NRBC	Number of nucleated red blood cells	10^9/L
NRBC	NRBC%	%
PLT	Platelets	10^9/L
MPV	Mean platelet volume	fL
PDW	Platelet distribution width	fL
PCT	Platelet hematocrit	%
P-LCR	Platelet -larger cell ratio	%
P-LCC	Platelet-large cell count	10^9/L
IPF	Immature platelet fraction	%
RET%	Reticulocyte percentage	%

RET#	umber of reticulocytes	10^9/L
IRF	Immature reticulocyte fraction	%
LFR	Low fluorescent reticulocyte ratio	%
MFR	Middle fluorescent reticulocyte ratio	%
HFR	High fluorescent reticulocyte ratio	%
RET-He	Reticulocyte hemoglobin concentration	pg

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Parameter	Parameter English Name	Default Unit
PLT-I	Number of platelets calculated from RBC/PLT channel	1X10^9/L
PLT-O	Number of platelets calculated from FCR channel	1X10^9/L
PLT-F	umber of platelets calculated from FCP channel	1X10^9/L
WBC-W	Number of WBC calculated from FCW channel	0.01X10^9/L
WBC-D	Number of WBC calculated from FCD channel	0.01X10^9/L
TNC	Total of nucleated cells	0.01X10^9/L
TNC-W	TNC calculated from FCW channel	0.01X10^9/L
TNC-D	Total of nucleated cells calculated from the FCD channel WBC + RBC	0.01X10^9/L
RBC-O	Number of RBC calculated from FCR channel	0.01X10^{12}/L
microR	Micro red blood cell ratio	0.1%
macroR	Macro red blood cell ratio	0.1%
NEUT#&	Number of granulocytes obtained by subtracting the number of IGs from the number of E Ts	0.01X10^9/L
NEUT%&	Ratio of E T number minus IG number to WBC count	0.1%
LYMP# &	Number of lymphocytes obtained by subtracting HFLC number from LYMPH number	0.01X10^9/L
LYMP%&	Ratio of LYMPH number minus HFLC number to WBC count	0.1%
HFLC#	Area count above LYMPH in the FCD scattergram	0.01X10^9/L
HFLC%	Ratio of area count above LYMPH in the FCD scattergram to WBC count	0.1%
BA-W#	Number of basophils calculated from FCW channel	0.01X10^9/L

BA-W%	Basophil percentage calculated from FCW channel	0.1%
BA-D#	umber of basophils calculated from FCD channel	0.01X10^9/L
BA-D%	Basophil percentage calculated from FCD channel	0.1%
RBC-He	Correlation between RBC-Y forward scattered light intensity of mature red blood cells and MCH to convert RBC-Y to pg unit	0.1pg
Delta-He	Calculated by the equation Delta-He = RET-He-RBC -He	0.1pg
RPI	Reticulocyte production index	/
H-IPF	Ratio of the number of platelets in the area of strong fluorescence intensity in the IPF region on the FCP scattergram to the total number of platelets	0.1%
IPF#	umber of platelets in the IPF region of the FCP scattergram	0.1X10^9/L
WBC-A	umber of WBC calculated from FCA channel	0.01X10^9/L
T C-A	T C calculated from FCA channel	0.01X10^9/L
HYPO-He	Ratio of particle number in the low-value area of the forward scattered light signal in the RBC mature red blood cell region of the FCR channel to RBC	0.1%
HYPER-He	Ratio of the number of particles in the high value are of the forward scattered light signal in the RBC mature red blood cell region of the FCR channel to RBC	0.1%
FCR- PP	Count in the PP region of the FCR scattergram	0.1X10^9/L
FCR-T C	Count in the T C region of the FCR scattergram	0.1X10^9/L
FRC#	Absolute number calculated from the number of particles in a specific area below the RBC particles on the FCR scattergram	0.1X10^9/L
FRC%	Percent calculated from the number of particles in a specific area below the RBC particles on the FCR scattergram	0.01%

5. Sampling Characteristics

- Sample size required for each analysis

Manual loading-WB 88μL
Manual loading-LW 88μL
Manual loading-PD 70μL
Manual loading-BF 200μL

6. Performance Index

- Blank count

Parameter	Blank count
WBC	≤ 0.10X109/L
RBC	≤ 0.02X1012/L
HGB	≤ 1g/L
PLT	≤ 5X109/L

- Linear

Parameter	Linear Range	Allowable Deviation Range	Linear Correlation Coefficient r
WBC	0.00-10.00X10^9/L	Not more than ±0.20X109/L	≥ 0.990
	10.01X10^9/L - 100.00X10^9/L	Not more than ±2%	
	100.01X10^9/L - 350.00X10^9/L	Not more than ±6%	
	350.01X10^9/L - 500.00X10^9/L	Not more than ±11%	
RBC	0- 1.50X10^{12}/L	Not more than ±0.03X1012/L	≥ 0.990
	1.51X10^{12}/L - 8.60X10^{12}/L	Not more than ±2.0%	
HGB	0.0- 100g/L	Not more than ±2g/L	≥ 0.990
	101g/L- 250g/L	Not more than ±2.0%	
	251g/L- 260g/L	Not more than ±5.0%	
PLT	0- 200X10^9/L	Not more than ±10X109/L	≥ 0.990
	201X10^9/L - 1000X10^9/L	Not more than ±5.0%	
	1001X10^9/L - 5000X10^9/L	Not more than ±6.0%	

■ Precision

Parameter	Detection Range	Precision %
WBC	≥ 3.50X10^{9}/L	≤2.5
RBC	≥ 3.50X10^{12}/L	≤ 1.5
HGB	110g/L - 180g/L	≤ 1.0
PLT	≥ 100X109/L	≤ 4.0
HCT or MCV	30% - 50%（HCT） or 80fL - 100fL（MCV）	≤1.5 HCT or ≤ 1.0 MCV

■ Accuracy

Parameter	Detection Range	Allowable Deviation Range %
WBC	3.5X10^{9}/L - 9.5X10^{9}/L	Not more than ±15.0
RBC	3.8X10^{12}/L - 5.8X10^{12}/L	Not more than ±6.0
HGB	115g/L - 175g/L	Not more than ±6.0
PLT	125X10^{9}/L - 350X10^{9}/L	Not more than ±20.0
HCT or MCV	35 - 50 HCT or 82fL - 100fL MCV	≤ 9.0 HCT or ≤ 7.0 MCV

■ White blood cell classification accuracy of 5-part differential analyzer

The measurement results of neutrophils, lymphocytes, monocytes, eosinophils and basophils obtained by the analyzer are within the allowable ranges of the results obtained according to the reference method 99 confidence interval .

Note: When the test result of the reference method is 0 and the test result of the analyzer is ≤ 1.0 , the test conclusion is qualified.

■ Carry-over rate

Parameter	Carry-o er rate requirement %
WBC	≤ 1.0
RBC	≤ 1.0
HGB	≤ 1.0
PLT	≤ 1.0

7.Dimensions and weight

	Length (mm)	Width (mm)	eight (mm)	Weight (kg)
Entire Instrument	≤ 670	≤ 760	≤ 810	≤ 76
Main unit of the analyzer	≤ 355	≤ 660	≤ 810	≤ 60

	Unit Module	Quantity	Unit
1	User Manual	1	Piece
2	Adapter	1	Set
3	European standard power cable	1	Piece
4	BS standard power cable	1	Piece
5	SA standard power cables.	1	Piece
6	Reagent lines	1	Set
7	Diluent lines	1	Set
8	Waste lines	1	Set
9	Sample rack HX 80	5	Piece
	Sample rack H X 8 5	5	Piece
	Sample rack HX 88	5	Piece
10	Sample loader optional	1	Piece
12	Scanner optional	1	Piece

EN ISO 13485:2016 CE

EU REP

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